

Harmonic Euler Sums and Lerch Zero Geometry

Entire Continuation, Infinite Direction Changes,
and an Optimal Exterior Barrier

Research continuation prepared for the ProveIt project

10 October 2026

Abstract

We continue the ProveIt polylogarithm research in three directions. First, the odd-denominator harmonic Euler sum $S(s) = \sum_{n \geq 0} (-1)^n H_n (2n+1)^{-s}$ extends to an entire function with exactly one simple real zero in each interval $(-2m-1, -2m)$, $m \geq 1$, and no other real zeros. A Fourier–beta identity and a quantitative variance bound prove the classification. We obtain exact negative-integer evaluations, a complex zero-free half-plane, monotone real-zero displacements, and all-orders zero asymptotics. Second, for the smallest Lerch zero on the diagonal $k = n - 1$, an exact inverse-function generating identity proves infinitely many positive and infinitely many negative initial velocities, for every positive lattice spacing. An explicitly labelled complex-singularity conjecture gives a proposed quantitative refinement. Third, we solve the incoming exterior-operator optimization problem: a cubic algebraic number determines the unique optimal constant coefficient and the least cutoff for the geometric sum. The strict pointwise problem has the same infimum, but does not attain it. Convergent coefficient formulas, two-scale cutoff expansions, exact rational certificates, independent numerical diagnostics, proposed manuscript corrections, and fourteen further research directions accompany the proofs. The already proved S_4 reduction is credited to the inspected baseline; the weight-seven S_6 reduction remains open.

Source baseline. VladimirReshetnikov/ProveIt
Commit: a0a90ef31877f98be437191c48b46f02d5456867.

Proof standard. Ordinary analytic and algebraic proofs, with reproducible finite certificates. Numerical illustrations are identified as diagnostics; no proof-assistant verification or exhaustive priority claim is implied.

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1 Introduction and the state of the problem

The ProveIt polylogarithm manuscript brings together exact cyclotomic identities, positive-kernel representations, and zero geometry of Stieltjes–Lerch derivatives. The most useful next step is to distinguish questions about a fixed integer weight from questions about an entire family. A special-value reduction asks which periods suffice to express one number. An order interpolation asks how those numbers fit into a holomorphic function. A zero-motion problem asks what remains uniform when two discrete indices change together. These questions interact, but a solution to one does not automatically solve the others.

This article develops three results beyond the inspected source baseline. The proofs are given in full, with classical analytic inputs either derived here or explicitly identified. The accompanying programs check finite identities and numerical conventions; the infinite conclusions do not rest on an integer-relation search or a scan of zeros.

1.1 The harmonic Euler sum as a function of its order

Let $H_0 = 0$ and $H_n = \sum_{j=1}^n j^{-1}$. The manuscript’s mixed Gaussian sums are the positive-integer values of

$$S(s) = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^s}, \quad \Re s > 0.$$

Our first main result is that this function extends to an entire function whose complete real zero set is

$$\{\sigma_m : m \geq 1\}, \quad -2m - 1 < \sigma_m < -2m,$$

with every zero simple. In particular $S(s) < 0$ for real $s \geq -2$. We also prove that S has no complex zero in $\Re s \geq 3$.

The continuation itself is a standard Mellin-regularization argument. The new work is the global zero classification. A differentiated Fourier–beta integral gives

$$S(-q) = \frac{cD(q)}{2} \{R(q) \cos(cq) - \sin(cq)\}, \quad c = \frac{\pi}{2},$$

where $D(q) > 0$ and R is a ratio of explicit Mellin moments. We prove $R(q) > 0$ and $0 < R'(q) < \pi/2$ for $q \geq 2$. This converts existence and uniqueness in every interval into a strict comparison with the tangent function. The derivative estimate follows from a one-dimensional variance inequality and elementary moment bounds. It covers the first interval as well as the asymptotic regime.

If $N = 2m + 1$, $\Lambda = \log(N/\pi) + \gamma$, and $\sigma_m = -N + \varepsilon_m$, then

$$\varepsilon_m = \frac{2}{\pi} \arctan \frac{\pi}{2\Lambda} - \frac{\frac{1}{2} - \frac{2}{\pi} \arctan(\pi/(2\Lambda))}{N(\Lambda^2 + \pi^2/4)} + O\left(\frac{1}{N^2\Lambda^2}\right).$$

We give both a complete expansion in inverse logarithms and a recursive construction of every correction in $1/N$. Exact negative-integer identities in generalized secant numbers supply an independent arithmetic description.

1.2 Diagonal Lerch zeros do not have an eventual common direction

For $n \geq 0$, $k \geq 1$, $h > 0$, and $0 \leq \rho \leq 1$, define

$$\mathcal{F}_{n,k}^{\rho,h}(a) = \sum_{m=0}^{\infty} \rho^m (-1)^{n+k} \frac{d^k}{da^k} \left(\frac{\log^n(a + mh)}{a + mh} \right), \quad a > 0.$$

The endpoint $\rho = 1$ is defined by this convergent derivative series. On the diagonal $k = n - 1$, the smallest elementary zero is $a = 1$. It continues to an analytic local zero branch $a_n^{(h)}(\rho)$. For $A = 1 + h$, put $v_n(A) = (a_n^{(h)})'(0)$. We prove the exact convergent generating identity

$$\sum_{n \geq 1} v_n(A) t^n = \log A - u_A(t), \quad e^{u_A(t)} = A + t u_A(t), \quad u_A(0) = \log A.$$

Here the $n = 1$ coefficient is an auxiliary term completing the generating function; the zero-branch assertion concerns $n \geq 2$.

For every real $A > 1$, infinitely many $v_n(A)$ are positive and infinitely many are negative. The proof uses Lagrange inversion and a positive-coefficient argument: the inverse u_A is analytic along the whole positive real axis and has unbounded logarithmic growth, which excludes eventually one-sign Taylor coefficients. This is compatible with a theorem that fixes n before letting k grow; it shows why that order of quantifiers matters.

1.3 An explicit solution to the exterior-operator problem

For $n \geq 2$, set

$$g_n(x) = x^{-2}(\log x)^{n-1}(n - \log x), \quad U_n(a, \rho) = \sum_{m \geq 0} \rho^m g_n(a + m).$$

The incoming *Lerch Global Phase* report asks for the best exterior cutoff obtainable by optimizing the positive constant in $\partial_a + \lambda$ [8]. Let r_n be the unique positive root of

$$r^3 + nr^2 - nr - n = 0.$$

The optimal coefficient is

$$\lambda_n = (2 - r_n)e^{-n-r_n}, \quad 1 < r_n < \frac{1 + \sqrt{5}}{2}.$$

We give a scalar equation uniquely defining the least cutoff B_n , prove

$$(\partial_a + \lambda_n)U_n(a, \rho) < 0 \quad (a \geq B_n, \ 0 < \rho \leq 1),$$

and prove that no smaller cutoff works uniformly over these parameters. At that cutoff the coefficient is unique. For the corresponding strict pointwise problem, the same cutoff is an infimum and is not attained: there is an unavoidable tangency in the negative tail. This boundary qualification is part of the solution.

The algebraic parameter r_n tends to the golden ratio. Its series in $1/n$ has exact radius of convergence $5/27$, and the cutoff has explicit exponentially small corrections in logarithmic coordinates. These are quantitative identities, rather than numerical fits.

1.4 Source baseline, attribution, and proof status

The source baseline is the repository commit

a0a90ef31877f98be437191c48b46f02d5456867,

dated 10 October 2026. The focused audit used all forty manuscript chapter files and the mathematical source material in the sixteen incoming archives listed in the accompanying provenance record. This is a focused research audit, not a claim to have independently re-proved every theorem in those archives.

At this baseline, the mixed Gaussian S_4 reduction is already proved by a convergent octahedral transformation and an exact finite certificate [2]. The S_2 companion, the product-matrix rank theorem, and several low-index Lerch phase results are also already established. The two displayed S_6 baskets are provably equivalent formulations of one remaining conjecture [3]. A frozen S_8 vector has already been disproved by an exact interval exclusion [5]. None of those earlier achievements is claimed as a result of this article.

Analytic continuation of harmonic Euler Dirichlet series has substantial precedent [11]; integer-weight Euler sums and their colored multiple-zeta representations are also established subjects [12]. Lagrange inversion, the Pringsheim principle, and the Brascamp–Lieb inequality are classical [14, 15, 13]. The claims of progress here are relative to the inspected manuscript and incoming reports. The literature check was targeted and does not establish global priority for every formula.

The results designated as theorems below have ordinary analytic or algebraic proofs. Exact rational computations provide additional finite certificates. Numerical quadrature and high-precision asymptotic comparisons are labelled diagnostics. The complex-singularity conjecture is kept separate from the proved infinite sign theorem, and S_6 remains open.

Question	Result in this article	Status
Real zeros of $S(s)$	Entire continuation; exactly one simple zero in each $(-2m-1, -2m)$; no other real zeros	Proved
Diagonal initial motion	Exact inverse generating function; infinitely many velocities of each sign for every $h > 0$	Proved
Best constant exterior operator	Unique optimal coefficient and least uniform summed cutoff; strict pointwise infimum identified	Proved
Quantitative diagonal oscillation	Candidate accessible conjugate singularities and coefficient asymptotics	Explicit conjecture
Mixed reduction for S_6	Existing candidate and its two-basket equivalence retained	Still open

Table 1: The principal proof-status distinctions.

2 Notation for the order interpolation

Write $H_0 = 0$, $H_n = \sum_{j=1}^n 1/j$ and

$$S(s) = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^s}, \quad \Re s > 0. \quad (2.1)$$

Use the positive real logarithm in the denominator, and let $\beta(s) = \sum_{n \geq 0} (-1)^n (2n+1)^{-s}$ denote the Dirichlet beta function and its entire continuation. Put

$$c = \frac{\pi}{2}, \quad G = \beta(2), \quad L(x) = \gamma + \Re \psi \left(\frac{1+ix}{2} \right) \quad (x \geq 0).$$

The gamma logarithmic derivative is ψ , and $\gamma = -\psi(1)$. In particular, $L(0) = -2 \log 2$.

Analytic continuation of alternating Euler sums and their values at negative integers belong to an established subject; see Boyadzhiev–Gadiyar–Padma [11]. Our continuation argument is an elementary instance of that method. The result developed here is the complete real zero classification for this specific odd-denominator harmonic interpolation, together with quantitative zero asymptotics. No exhaustive literature priority claim is made.

3 Entire continuation and a zero-free real ray

Define, for $t \geq 0$,

$$K(t) = \frac{e^{-t} \log(1 + e^{-2t})}{1 + e^{-2t}} = \frac{\log(2 \cosh t) - t}{2 \cosh t}. \quad (3.1)$$

The second equality holds on the real line; near zero both sides have the same analytic germ. The generating function of the harmonic numbers gives

$$-K(t) = \sum_{n \geq 0} (-1)^n H_n e^{-(2n+1)t} \quad (t > 0).$$

Consequently, initially by absolute convergence for $\Re s > 1$,

$$S(s) = -\frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} K(t) dt. \quad (3.2)$$

The original series is locally uniformly convergent for $\Re s > 0$: summation by parts applies because $H_n(2n+1)^{-s} \rightarrow 0$ and the successive differences are $O_C(n^{-1-\sigma} \log n)$ on every compact set $C \subset \{\Re s \geq \sigma > 0\}$. Thus (3.2) represents (2.1) throughout that half-plane.

Proposition 3.1. *The function S extends to an entire function, real on the real axis. For every integer $j \geq 0$,*

$$S(-j) = (-1)^{j+1} K^{(j)}(0). \quad (3.3)$$

Moreover $S(s) < 0$ for every real $s \geq -2$.

Proof. The function K is analytic near zero and is $O(e^{-3t})$ at positive infinity, as are each of its fixed-order derivatives. Subtract its Taylor polynomial on $[0, 1]$. The Mellin integral then continues meromorphically across any finite part of the plane, with possible simple poles at $s = -j$ and residues $K^{(j)}(0)/j!$. The reciprocal gamma function has a simple zero there with leading coefficient $(-1)^j j!$. This cancels every pole and gives (3.3). The construction uses real Taylor coefficients and a real kernel, proving the reality assertion.

For the sign assertion we show $K''(t) > 0$ on $[0, \infty)$. Set $y = e^{-2t} \in (0, 1]$. Direct differentiation gives

$$K''(t) = \frac{\sqrt{y}}{(1+y)^3} \left((y^2 - 6y + 1) \log(1+y) - 4y^2 + 8y \right). \quad (3.4)$$

If $y^2 - 6y + 1 \geq 0$, the bracket is positive. Otherwise $\log(1+y) < y$, so the bracket is strictly greater than

$$(y^2 - 6y + 1)y - 4y^2 + 8y = y(y-1)(y-9) \geq 0.$$

Two integrations by parts in (3.2) give

$$S(s) = -\frac{1}{\Gamma(s+2)} \int_0^\infty t^{s+1} K''(t) dt, \quad \Re s > -2. \quad (3.5)$$

The integrations are first justified for $\Re s > 0$; analyticity then extends the identity to the stated half-plane. For real $s > -2$ all factors in the integral and the gamma denominator are positive, so $S(s) < 0$. At the endpoint, $S(-2) = -K''(0) = (\log 2 - 1)/2 < 0$. $\square$

Corollary 3.2 (A complex zero-free half-plane). *For every $s \in \mathbb{C}$ with $\Re s \geq 3$,*

$$|3^s S(s) + 1| < \frac{5}{6}.$$

In particular S has no zero in that half-plane.

Proof. The $n = 1$ term of $3^s S(s)$ is -1 , and absolute convergence gives

$$|3^s S(s) + 1| \leq 27 \sum_{n=2}^{\infty} \frac{H_n}{(2n+1)^3}.$$

For $n \geq 6$, use $H_n \leq 1 + \log n$ and $(2n+1)^3 > 8n^3$. The function $(1 + \log x)x^{-3}$ decreases for $x \geq 5$, so the tail from $n = 6$ is at most $(3 + 2 \log 5)/800 < 7/800$. Therefore

$$\sum_{n=2}^{\infty} \frac{H_n}{(2n+1)^3} < \sum_{n=2}^5 \frac{H_n}{(2n+1)^3} + \frac{7}{800} = \frac{122481517493}{3993750684000}.$$

Multiplication by 27 gives a rational number strictly below $5/6$. $\square$

3.1 Exact values at the negative integers

Define the generalized secant numbers by

$$(\sec z)^\alpha = \sum_{m \geq 0} E_{2m}(\alpha) \frac{z^{2m}}{(2m)!}, \quad E_{2m} = E_{2m}(1) > 0.$$

A prime on $E_{2m}(\alpha)$ will mean differentiation with respect to α . Comparison of the even and odd parts of (3.1) gives the following exact formulas.

Corollary 3.3. For $m \geq 0$,

$$S(-2m) = \frac{(-1)^m}{2} (E'_{2m}(1) - E_{2m} \log 2), \quad (3.6)$$

$$S(-2m-1) = \frac{(-1)^{m+1}}{2} (2m+1)E_{2m}. \quad (3.7)$$

For $m \geq 1$, $(-1)^m S(-2m) > 0$.

Proof. The even part of K is $\frac{1}{2} \operatorname{sech} t [\log 2 + \log \cosh t]$ and its odd part is $-\frac{1}{2} t \operatorname{sech} t$. Use $\partial_\alpha (\operatorname{sech} t)^\alpha|_{\alpha=1} = -\operatorname{sech} t \log \cosh t$ and $\operatorname{sech} t = \sec(it)$ in (3.3). This proves both identities. The Taylor coefficients of $\log \sec z$ in positive even degrees are positive: its derivative is $\tan z$, whose positive coefficients follow recursively from $u' = 1 + u^2$, $u(0) = 0$. Therefore, when $m \geq 1$, $E_{2m}(\alpha)$ is a nonzero polynomial in positive powers of α with nonnegative coefficients. It follows that $E'_{2m}(1) \geq E_{2m}$, proving the sign because $\log 2 < 1$. $\square$

For example,

$$\begin{array}{c|cccccccc} s & 0 & -1 & -2 & -3 & -4 & -5 & -6 \\ \hline S(s) & -\frac{\log 2}{2} & -\frac{1}{2} & \frac{\log 2 - 1}{2} & \frac{3}{2} & 4 - \frac{5}{2} \log 2 & -\frac{25}{2} & \frac{61 \log 2 - 121}{2} \end{array}$$

The sign change between $-2m-1$ and $-2m$ already proves the existence of at least one zero in each interval. It gives neither uniqueness nor exclusion of other real zeros; those require the next argument.

4 A Fourier–beta identity

For real $\alpha > 0$ define

$$B_\alpha(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{(2 \cosh t)^\alpha} dt, \quad \Re s > 0.$$

In particular $B_1(s) = \beta(s)$. From the kernels, or from the binomial series and $(\alpha)_n$ differentiated at $\alpha = 1$,

$$S(s) = s\beta(s+1) + \partial_\alpha B_\alpha(s)|_{\alpha=1}. \quad (4.1)$$

This identity extends wherever its two sides are continued.

Lemma 4.1. For $\Re s < 1$,

$$B_\alpha(s) = \frac{\cos(\pi s/2)}{2\pi\Gamma(\alpha)} \int_0^\infty x^{-s} \Gamma\left(\frac{\alpha + ix}{2}\right) \Gamma\left(\frac{\alpha - ix}{2}\right) dx. \quad (4.2)$$

On the left this means analytic continuation. Differentiation with respect to α is valid locally uniformly near $\alpha = 1$.

Proof. The substitution $u = e^{2t}$ and the beta integral show

$$\int_{-\infty}^\infty \frac{e^{ixt}}{(2 \cosh t)^\alpha} dt = \frac{\Gamma((\alpha + ix)/2) \Gamma((\alpha - ix)/2)}{2\Gamma(\alpha)}.$$

Fourier inversion and the Mellin transform of the cosine prove (4.2) first in $0 < \Re s < 1$. For a direct justification of the interchange, insert $e^{-\varepsilon t}$ before the Mellin integral, integrate in t , and then send $\varepsilon \downarrow 0$. The resulting kernels are bounded near zero by a constant times $x^{-\Re s}$, and the gamma product decays exponentially at infinity. Dominated convergence applies. The right side is holomorphic in $\Re s < 1$, so it supplies the claimed continuation. Differentiation in α only introduces digamma factors growing logarithmically at infinity; the same domination applies on a compact positive α -interval. $\square$

For real $q > 0$ put

$$J(q) = \int_0^\infty x^q \operatorname{sech}(cx) dx, \quad (4.3)$$

$$A(q) = \int_0^\infty x^q \operatorname{sech}(cx) L(x) dx, \quad (4.4)$$

$$D(q) = \int_0^\infty x^q \operatorname{sech}(cx) \tanh(cx) dx, \quad (4.5)$$

$$R(q) = \frac{A(q)}{cD(q)}. \quad (4.6)$$

The letter D here denotes a positive integral, not differentiation.

Proposition 4.2. For $q > 0$,

$$S(-q) = \frac{cD(q)}{2} (R(q) \cos(cq) - \sin(cq)). \quad (4.7)$$

Moreover,

$$J(q) = 2\Gamma(q+1)c^{-q-1}\beta(q+1), \quad D(q) = \frac{q}{c}J(q-1). \quad (4.8)$$

Proof. At $\alpha = 1$ the reflection formula gives $\Gamma((1+ix)/2)\Gamma((1-ix)/2) = \pi \operatorname{sech}(cx)$. The logarithmic α derivative of the full gamma quotient in (4.2) is $L(x)$. Thus the derivative term in (4.1) at $s = -q$ is $\frac{1}{2} \cos(cq)A(q)$. Applying the same Fourier formula to $\beta(1-q)$ gives $\beta(1-q) = \frac{1}{2} \sin(cq)J(q-1)$. Since $D(q) = qJ(q-1)/c$ by integration by parts, the remaining term is $-\frac{1}{2}c \sin(cq)D(q)$. This proves (4.7). The expansion $\operatorname{sech}(cx) = 2 \sum_{n \geq 0} (-1)^n e^{-(2n+1)cx}$ gives the stated formula for $J(q)$ for $q > 0$ by absolute integration. The integration-by-parts identity for D is valid for every $q > 0$. $\square$

5 The derivative bound that forces uniqueness

The next estimate is the central analytic input. Its numerical constant is deliberately coarse; it is small enough for the tangent comparison and needs no numerical certification.

Lemma 5.1. *Let $r(x) = L(x) \coth(cx)/c$ for $x > 0$. Then*

$$0 < xr'(x) \leq a + \frac{b}{x}, \quad a = \frac{2}{c} = \frac{4}{\pi}, \quad b = \frac{2 + 2 \log 2}{c^2}. \quad (5.1)$$

Proof. The digamma series gives

$$L(x) = -2 \log 2 + 2 \sum_{\substack{k \geq 1 \\ k \text{ odd}}} \frac{x^2}{k(k^2 + x^2)}, \quad (5.2)$$

$$L'(x) = 4x \sum_{\substack{k \geq 1 \\ k \text{ odd}}} \frac{k}{(k^2 + x^2)^2} > 0. \quad (5.3)$$

These series converge locally uniformly in x . Also

$$L'(x) = \int_0^\infty \frac{u \sin(xu)}{\sinh u} du.$$

Writing $h(u) = u/\sinh u$, integration by parts yields

$$xL'(x) = 1 + \int_0^\infty h'(u) \cos(xu) du.$$

Here h decreases from 1 to 0, so $\int_0^\infty |h'(u)| du = 1$ and $0 < xL'(x) \leq 2$.

Now

$$r'(x) = \frac{L'(x)}{c} \coth(cx) - L(x) \operatorname{csch}^2(cx).$$

This is positive whenever $L(x) \leq 0$. If $L(x) > 0$, comparison term by term in (5.2)–(5.3) gives

$$L(x) < L(x) + 2 \log 2 \leq \frac{1 + x^2}{2} xL'(x).$$

It follows that

$$r'(x) > \frac{L'(x)}{c} \coth(cx) \left(1 - \frac{cx(1 + x^2)}{\sinh(2cx)} \right) > 0.$$

The last inequality follows from $\sinh(\pi x) \geq \pi x + (\pi x)^3/6 > cx(1 + x^2)$. Finally $L(x) \geq -2 \log 2$, $\coth y \leq 1 + 1/y$ and $\sinh y \geq y$ give

$$xr'(x) \leq \frac{2}{c} \left(1 + \frac{1}{cx} \right) + \frac{2 \log 2}{c^2 x},$$

which is (5.1). □

For completeness we give the needed one-dimensional form of the Brascamp–Lieb variance inequality [13].

Lemma 5.2. *For a probability density proportional to $e^{-V(x)}$ on an interval, with $V''(x) > 0$,*

$$\operatorname{Var}(f) \leq \mathbb{E} \frac{f'(x)^2}{V''(x)}.$$

Consequently $|\operatorname{Cov}(f, g)|^2 \leq \mathbb{E}(f'^2/V'') \mathbb{E}(g'^2/V'')$ whenever the displayed quantities exist.

Proof. It suffices first to work on a compact interval $[a, b]$ and subtract the mean of f . Put $u(x) = e^{V(x)} \int_a^x f(t) e^{-V(t)} dt$. Then $u(a) = u(b) = 0$ and $f = u' - V'u$. Integration by parts gives

$$\mathbb{E}f^2 = -\mathbb{E}(f'u), \quad \mathbb{E}f^2 = \mathbb{E}u'^2 + \mathbb{E}(V''u^2).$$

Cauchy–Schwarz with weights V'' implies $(\mathbb{E}f^2)^2 \leq \mathbb{E}(f'^2/V'')\mathbb{E}(V''u^2) \leq \mathbb{E}(f'^2/V'')\mathbb{E}f^2$. Division proves the variance bound. Exhaust the original interval by compact intervals and use truncation and convergence of the stated integrals. The covariance assertion then follows from Cauchy–Schwarz and the two variance bounds. $\square$

Theorem 5.3. *For every real $q \geq 2$,*

$$R(q) > 0, \quad 0 < R'(q) < \frac{1}{q} \sqrt{\frac{181}{27}} < \frac{\pi}{2}. \quad (5.4)$$

Proof. Give $(0, \infty)$ the probability density

$$d\nu_q(x) = \frac{x^q \operatorname{sech}(cx) \tanh(cx)}{D(q)} dx.$$

Then $R(q) = \mathbb{E}_{\nu_q} r(X)$ and differentiation in q gives $R'(q) = \operatorname{Cov}_{\nu_q}(r(X), \log X) > 0$, because both functions are strictly increasing and the density has full support. Differentiation is justified by absolute integrability of the additional logarithm. Since $S(-2) = (\log 2 - 1)/2$, (4.7) gives $A(2) = 1 - \log 2 > 0$, whence $R(2) > 0$ and $R(q) > 0$ for all $q \geq 2$.

The negative logarithm of the density, up to a constant, is

$$V_q(x) = -q \log x + \log \cosh(cx) - \log \tanh(cx).$$

Its second derivative is

$$V_q''(x) = \frac{q}{x^2} + c^2 \operatorname{sech}^2(cx) + 4c^2 \frac{\cosh(2cx)}{\sinh^2(2cx)} \geq \frac{q}{x^2}.$$

Lemma 5.2, applied to r and $\log x$, therefore yields

$$R'(q) \leq \frac{1}{q} \sqrt{\mathbb{E}_{\nu_q}(X r'(X))^2}. \quad (5.5)$$

There is no endpoint assumption hidden here: as $x \downarrow 0$, $r(x) = O(x^{-1})$, $xr'(x) = O(x^{-1})$, and the density is $O(x^{q+1})$. All second moments and energy integrals used in the lemma are finite for $q \geq 2$. The tails are exponentially integrable.

For $j = 1, 2$, $\mathbb{E}_{\nu_q} X^{-j}$ decreases with q by the same covariance differentiation, now applied to a decreasing function and $\log x$. At $q = 2$ the two inverse moments coincide. Indeed,

$$D(0) = \frac{1}{c}, \quad D(1) = \frac{1}{c}, \quad D(2) = \frac{4G}{c^3},$$

so

$$\mathbb{E}_{\nu_q} X^{-1}, \mathbb{E}_{\nu_q} X^{-2} \leq \mu := \frac{\pi^2}{16G} < \frac{3}{4}.$$

Here $G > 1 - 1/9 = 8/9$ by the alternating-series estimate and $\pi^2 < 10$, so even $\mu < 45/64$ suffices. The elementary inequalities $\pi > 3$ and $\log 2 < 3/4$ give $a < 4/3$ and $b < 14/9$ in Lemma 5.1. Consequently

$$\mathbb{E}_{\nu_q}(X r'(X))^2 \leq a^2 + 2ab\mu + b^2\mu < \frac{16}{9} + \frac{28}{9} + \frac{49}{27} = \frac{181}{27}.$$

This proves the first derivative bound. For $q \geq 2$ its square is at most $181/108 < 9/4 < c^2$, proving the final strict inequality. $\square$

6 Complete classification of the real zeros

Theorem 6.1. *For each integer $m \geq 1$, S has exactly one real zero*

$$\sigma_m \in (-2m - 1, -2m).$$

Every such zero is simple. These are all the real zeros of S . The displacements $\varepsilon_m = \sigma_m + 2m + 1$ strictly decrease with m . In particular, $\sigma_m - \sigma_{m+1} > 2$.

Proof. There are no zeros on $[-2, \infty)$ by Proposition 3.1. Write $q = -s > 2$. On $(2m, 2m + 1)$, equation (4.7) is equivalent to

$$\tan(c(q - 2m)) = R(q).$$

The left side increases from 0 to $+\infty$. The right side is positive and finite, and the derivative of their difference is $c \sec^2(c(q - 2m)) - R'(q) > 0$ by Theorem 5.3. There is therefore exactly one solution. At that solution the same strict inequality and $\cos(cq) \neq 0$ show that the derivative of $S(-q)$ is nonzero, proving simplicity.

On $(2m + 1, 2m + 2)$, $\cos(cq)$ and $-\sin(cq)$ have the same sign. Since $R(q) > 0$ and $D(q) > 0$, (4.7) cannot vanish. At the integer endpoints one of the sine and cosine is zero and the other is nonzero, while $R, D > 0$. This exhausts $q > 2$.

For the monotonicity write $q_m = 2m + \delta_m$, where $\delta_m \in (0, 1)$ and $\tan(c\delta_m) = R(2m + \delta_m)$. Increasing m strictly increases the right side at every fixed δ . Since the difference between the left and right sides is strictly increasing in δ , its unique zero moves to the right. Thus $\delta_{m+1} > \delta_m$, or $\varepsilon_{m+1} < \varepsilon_m$, as claimed. $\square$

7 All-orders asymptotics of the zero displacement

For $k \geq 1$ define

$$d_k = \frac{(-1)^{k+1} 2^{2k} B_{2k}(1/2)}{2k}, \quad d_1 = -\frac{1}{6}, \quad d_2 = -\frac{7}{60}, \quad d_3 = -\frac{31}{126}.$$

Here $B_n(x)$ is the Bernoulli polynomial. Let $q^j = q(q - 1) \cdots (q - j + 1)$.

Lemma 7.1. *For every fixed integer $K \geq 1$, as real $q \rightarrow +\infty$,*

$$cR(q) = \psi(q + 1) + \gamma - \log \pi + \sum_{k=1}^{K-1} \frac{d_k c^{2k}}{q^{2k}} + O_K(q^{-2K}). \quad (7.1)$$

In particular,

$$cR(q) = \log(q/\pi) + \gamma + \frac{1}{2q} - \frac{2 + \pi^2}{24q^2} + O(q^{-3}). \quad (7.2)$$

Proof. The ordinary sectorial digamma expansion [17, §5.11], applied on the vertical line with real part $1/2$, gives

$$L(x) = \gamma + \log(x/2) + \sum_{k=1}^{K-1} d_k x^{-2k} + O_K(x^{-2K}) \quad (x \rightarrow \infty).$$

The coefficient formula follows either from the shifted Bernoulli expansion of $\psi(z + 1/2)$ or by expanding the ordinary formula in $1/(ix/2)$. The error may be bounded globally by $C_K x^{-2K}$ for all $x > 0$: on $(0, 1]$ the actual function L is bounded, and each subtracted power and logarithm is bounded by a constant times x^{-2K} . Hence for $q > 2K - 1$,

$$A(q) = J'(q) + (\gamma - \log 2)J(q) + \sum_{k=1}^{K-1} d_k J(q - 2k) + O_K(J(q - 2K)).$$

Use (4.8). The first quotient, multiplied by c , is

$$\frac{\beta(q+1)}{\beta(q)} \left(\psi(q+1) + \gamma - \log \pi + \frac{\beta'(q+1)}{\beta(q+1)} \right).$$

The k th remaining quotient is $d_k c^{2k} \beta(q-2k+1)/(q^{2k} \beta(q))$. For fixed shifts, the beta ratios are $1 + O_K(3^{-q})$, and the logarithmic derivative is $O_K(3^{-q})$, directly from their absolutely convergent Dirichlet series. The integrated error divided by $qJ(q-1)/c$ is $O_K(q^{-2K})$. This proves (7.1); expanding $\psi(q+1)$ gives (7.2). $\square$

Theorem 7.2. *Write*

$$N = 2m + 1, \quad \sigma_m = -N + \varepsilon_m, \quad \Lambda = \log(N/\pi) + \gamma, \quad Q = \Lambda^2 + c^2, \quad e_0 = \frac{1}{c} \arctan \frac{c}{\Lambda}.$$

Then $0 < \varepsilon_m < 1$ and

$$\varepsilon_m = e_0 - \frac{\frac{1}{2} - e_0}{NQ} + O\left(\frac{1}{N^2 \Lambda^2}\right) \quad (m \rightarrow \infty). \quad (7.3)$$

For every fixed $K \geq 1$,

$$\varepsilon_m = \sum_{j=0}^{K-1} \frac{(-1)^j c^{2j}}{(2j+1) \Lambda^{2j+1}} + O_K(\Lambda^{-2K-1}). \quad (7.4)$$

Thus the first three logarithmic terms are $\Lambda^{-1} - \pi^2/(12\Lambda^3) + \pi^4/(80\Lambda^5)$. The consecutive-zero gaps satisfy

$$\sigma_m - \sigma_{m+1} = 2 + \frac{2}{N(\Lambda^2 + c^2)} + O\left(\frac{1}{N^2 \Lambda^2}\right). \quad (7.5)$$

Proof. The exact zero equation is

$$c \cot(c\varepsilon_m) = cR(N - \varepsilon_m). \quad (7.6)$$

Lemma 7.1, uniformly for $0 \leq \varepsilon \leq 1$, gives

$$cR(N - \varepsilon) = \Lambda + \frac{\frac{1}{2} - \varepsilon}{N} + O(N^{-2}).$$

The solution of $c \cot(c\varepsilon_0) = \Lambda$ is the stated e_0 for all sufficiently large N . The right side of the preceding estimate shows that the left side at ε_m differs from Λ by $O(N^{-1})$. Its inverse has derivative $-1/(\Lambda^2 + c^2)$ at Λ , so $\varepsilon_m - e_0 = O(N^{-1} \Lambda^{-2})$. Taylor expansion of the left side about e_0 , whose first derivative is $-Q$ and second derivative is $2Q\Lambda$, now yields

$$-Q(\varepsilon_m - e_0) = \frac{\frac{1}{2} - e_0}{N} + O(N^{-2}).$$

The quadratic error is $O(N^{-2}/\Lambda)$ and is absorbed in the displayed bound; the displacement inside the right side is even smaller. Division by Q proves (7.3). Finally expand the arctangent. Its Taylor remainder gives the error in (7.4), while the inverse-index correction is smaller than every fixed power of $1/\Lambda$. For (7.5), regard the displayed approximation as a smooth function of real N . Its leading term has derivative $e'_0(N) = -1/(N(\Lambda^2 + c^2))$, and the derivative of the explicit inverse-index correction is $O(N^{-2} \Lambda^{-2})$. Take the difference at N and $N+2$ in (7.3); its two remainder values have the same $O(N^{-2} \Lambda^{-2})$ bound. $\square$

An algorithm for every inverse-index correction. Equation (7.1) supplies a full expansion in $1/q$. Substituting $\varepsilon = e_0 + \sum_{j \geq 1} a_j(\Lambda) N^{-j}$ into (7.6) and comparing coefficients determines a_j recursively: the coefficient of the new unknown is always $-Q \neq 0$. The functions a_j belong to $\mathbb{Q}[c^2, \Lambda, e_0, Q^{-1}]$, with Bernoulli numbers as rational coefficients, and $a_j = O_j(\Lambda^{-2})$. For each fixed $J \geq 1$ the truncation through $a_{J-1} N^{1-J}$ has error $O_J(N^{-J} \Lambda^{-2})$. To justify this statement, first use the full remainder in (7.1) to bound the residual after any fixed formal substitution by $O_J(N^{-J})$. Derivatives of $c \cot(c\varepsilon)$ at e_0 are polynomials in Λ, c^2 of degree at most the derivative order plus one. Induction then gives $a_j = O_j(\Lambda^{-2})$ and controls all Taylor remainders. The derivative of the exact implicit equation has magnitude comparable to Q , by (5.4) and $\varepsilon \asymp 1/\Lambda$. The mean value theorem divides the residual by Q , proving the stated error. This is an asymptotic recursion; no convergence of the resulting infinite formal series is asserted.

8 A diagonal zero branch and infinitely many initial directions

The fixed-index monotonicity theorem permits its derivative-order threshold to depend on the Laurent index. That dependence is essential. We prove that on the diagonal $k = n - 1$, the smallest zero issuing from the elementary endpoint moves initially to the right for infinitely many n and initially to the left for infinitely many n . An exact inverse-function identity organizes all these velocities. The result concerns the central elementary zero at $a = 1$, which is separate from the branches near positive Bessel zeros in the earlier adjacent-index expansion.

8.1 The shifted lattice and its elementary endpoint

It is useful to allow an arbitrary lattice spacing $h > 0$. For integers $n \geq 0, k \geq 1$, define

$$f_{n,k}(a) = (-1)^{n+k} \frac{d^k}{da^k} \left(\frac{\log^n a}{a} \right), \quad a > 0, \quad (8.1)$$

and

$$\mathcal{F}_{n,k}^{\rho,h}(a) = \sum_{m=0}^{\infty} \rho^m f_{n,k}(a + mh), \quad 0 \leq \rho \leq 1. \quad (8.2)$$

At $h = 1$ this is the normalized Stieltjes–Lerch derivative family used in the manuscript. For $\rho < 1$ it is $(-1)^{n+k}$ times the k th derivative of the corresponding logarithmic Lerch series. At $\rho = 1$ we use (8.2) itself; the undifferentiated logarithmic series need not converge. The derivative series converges absolutely because

$$f_{n,k}(x) = O_{n,k}(x^{-k-1}(1 + \log^n x)) \quad (x \rightarrow \infty).$$

The convergence is locally uniform in $a > 0$, together with any fixed number of a -derivatives. Near $\rho = 0$ the series is jointly holomorphic in a and ρ , using the logarithm analytic near the positive axis.

The elementary polynomial is

$$P_{n,k}(x) = n! [z^n] e^{xz} \prod_{j=1}^k \left(1 + \frac{z}{j} \right). \quad (8.3)$$

Differentiating a^{-1-z} and comparing coefficients gives

$$f_{n,k}(a) = k! a^{-k-1} P_{n,k}(-\log a). \quad (8.4)$$

Lemma 8.1 (The central diagonal zero). *For every $n \geq 2$, $f_{n,n-1}$ has exactly n simple positive zeros. Its smallest zero is 1, and*

$$f'_{n,n-1}(1) = -n!. \quad (8.5)$$

For every $h > 0$ there is an $\epsilon_{n,h} > 0$ such that $\mathcal{F}_{n,n-1}^{\rho,h}$ has exactly n simple positive zeros for $0 \leq \rho < \epsilon_{n,h}$. Its smallest zero is a real analytic branch $a_n^{(h)}(\rho)$ with $a_n^{(h)}(0) = 1$.

Proof. The polynomial in (8.3) is obtained from x^n by applying k operators $1 + \partial_x/j$. If a real-rooted polynomial p has distinct roots r_i of multiplicities m_i , then away from these roots the zeros of $p + cp'$, $c > 0$, solve

$$\frac{p'(x)}{p(x)} = \sum_i \frac{m_i}{x - r_i} = -\frac{1}{c}.$$

The left side is strictly decreasing on each complementary interval. There is one new simple root between consecutive distinct roots and one to the left of the smallest root. An inherited root of multiplicity $m_i \geq 2$ has multiplicity $m_i - 1$. Starting from x^n , after $n - 1$ operators all n roots are simple; zero is the largest and the others are negative. Equation (8.4) proves the first claim.

Since $n + (n - 1)$ is odd,

$$f_{n,n-1}(a) = -\frac{d^{n-1}}{da^{n-1}} \left(\frac{\log^n a}{a} \right).$$

The expansion $\log^n a/a = (a - 1)^n + O((a - 1)^{n+1})$ proves (8.5). The analytic implicit function theorem therefore continues this zero, as well as each of the other simple zeros.

For completeness, no additional positive zeros enter from an endpoint. For sufficiently large a , every term $f_{n,n-1}(a + mh)$ has the same strict sign, because its argument exceeds the largest elementary zero. As $a \downarrow 0$, the $m = 0$ term tends to $+\infty$, while the sum over $m \geq 1$ is bounded uniformly for a in a small positive interval and $0 \leq \rho \leq 1$. Thus there are no zeros sufficiently close to zero. On the remaining compact set outside small neighborhoods of the elementary zeros, uniform convergence to $f_{n,n-1}$ as $\rho \downarrow 0$ excludes zeros. Inside each neighborhood, uniform convergence of the first a -derivative preserves its strict sign and hence gives exactly one simple zero. The branch issuing from 1 is consequently the smallest. $\square$

8.2 An inverse function generates every diagonal velocity

Let $\left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right]$ denote the unsigned Stirling numbers of the first kind, fixed by

$$\prod_{r=0}^{n-1} (z + r) = \sum_{j=0}^n \left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right] z^j.$$

Theorem 8.2 (Exact velocity identity). *For $A > 1$ and $n \geq 1$, put*

$$v_n(A) = -\frac{1}{n!} \frac{d^{n-1}}{dx^{n-1}} \left(\frac{\log^n x}{x} \right) \Big|_{x=A} \quad (8.6)$$

$$= A^{-n} \sum_{j=1}^n \left[\begin{smallmatrix} n \\ n - j + 1 \end{smallmatrix} \right] \frac{(-\log A)^j}{j!}. \quad (8.7)$$

For the lattice spacing $h = A - 1$, the smallest diagonal branch satisfies

$$(a_n^{(h)})'(0) = v_n(A), \quad n \geq 2. \quad (8.8)$$

There is a unique analytic germ $u_A(t)$ at zero satisfying

$$e^{u_A(t)} = A + tu_A(t), \quad u_A(0) = \log A. \quad (8.9)$$

Its coefficients give the convergent ordinary generating identity

$$V_A(t) := \sum_{n=1}^{\infty} v_n(A)t^n = \log A - u_A(t) \quad (8.10)$$

for all sufficiently small complex t .

Proof. Differentiate the zero equation at $\rho = 0$. The shifted series and (8.5) give

$$(a_n^{(h)})'(0) = -\frac{f_{n,n-1}(1+h)}{f'_{n,n-1}(1)} = \frac{f_{n,n-1}(A)}{n!},$$

which is (8.6). The Stirling identity implies

$$P_{n,n-1}(x) = n \sum_{j=1}^n \left[\begin{matrix} n \\ n-j+1 \end{matrix} \right] \frac{x^j}{j!}.$$

Together with (8.4), this proves (8.7). The same finite identity holds at $n = 1$ by direct calculation; that term is included for the generating function.

Let $y(t)$ solve $y = A + t \log y$, with $y(0) = A$. The analytic implicit function theorem supplies a unique germ. The Lagrange–Bürmann formula with outer function $\log y$ gives

$$[t^n] \log y(t) = \frac{1}{n!} \frac{d^{n-1}}{dx^{n-1}} \left(\frac{\log^n x}{x} \right) \Big|_{x=A}.$$

This is the standard analytic form of Lagrange inversion [14]. Set $u_A = \log y$. Its equation is (8.9), and coefficient comparison proves (8.10). $\square$

Corollary 8.3 (An exact quadratic recurrence). *Write $L = \log A$ and $p_n(L) = A^n v_n(A)$. Then $p_1(L) = -L$, and for $n \geq 1$,*

$$(n+1)p_{n+1}(L) = (n+1-nL)p_n(L) + \frac{n}{2} \sum_{j=1}^{n-1} p_j(L)p_{n-j}(L). \quad (8.11)$$

Each p_n is a rational polynomial of degree n , with constant term zero, linear coefficient -1 , and leading coefficient $(-1)^n/n$.

Proof. The generating identity is equivalent to

$$Ae^{-V_A(t)} = A + t(L - V_A(t)).$$

Differentiation and elimination of the exponential yield

$$(A + t(L - 1 - V_A(t)))V'_A(t) = V_A(t) - L. \quad (8.12)$$

The constant coefficient gives $v_1 = -L/A$. At degree $n \geq 1$, the convolution is

$$\sum_{j=1}^{n-1} (n-j)v_j v_{n-j} = \frac{n}{2} \sum_{j=1}^{n-1} v_j v_{n-j}.$$

Multiplication by A^n gives (8.11). The coefficient statements follow directly from (8.7), using $\begin{bmatrix} n \\ n \end{bmatrix} = 1$ and $\begin{bmatrix} n \\ 1 \end{bmatrix} = (n-1)!$. $\square$

Corollary 8.4 (Nonvanishing at algebraic lattice endpoints). *If $A > 1$ is algebraic, then $v_n(A) \neq 0$ for every $n \geq 1$.*

Proof. The Hermite–Lindemann theorem implies that $\log A$ is transcendental: it is nonzero, and its exponential is algebraic. Each p_n in Corollary 8.3 is a nonzero rational polynomial. Thus $p_n(\log A) \neq 0$, and $v_n(A) = A^{-n}p_n(\log A) \neq 0$. $\square$

8.3 A sign theorem without an unproved singularity analysis

Lemma 8.5 (Positive-axis continuation and coefficient signs). *Let g be a nonpolynomial analytic germ at zero with real Taylor coefficients. Suppose it admits coherent holomorphic continuation to a neighborhood of every point of $[0, \infty)$ and has at most polynomial growth on the positive real axis. Then its Taylor coefficients include infinitely many positive and infinitely many negative terms.*

Proof. Suppose the coefficients have one sign from some index N onward. Subtract the Taylor polynomial of degree $N - 1$ and, if needed, multiply by -1 . We obtain a nonpolynomial germ

$$H(t) = \sum_{n \geq N} b_n t^n, \quad b_n \geq 0,$$

with the same continuation and growth properties.

Here is the needed positive-coefficient argument in full. If the radius of convergence were a finite $R > 0$, analyticity near the positive point R would imply, for every $j \geq 0$,

$$H^{(j)}(R) = \sum_{n \geq j} (n)_j b_n R^{n-j} \geq 0.$$

Indeed the equality holds first at $0 < t < R$; monotone convergence of its nonnegative terms and continuity of $H^{(j)}$ at R give the displayed formula. Choose $\delta > 0$ within the Taylor radius at R . That Taylor series converges absolutely, so Tonelli's theorem and the binomial theorem give

$$\sum_{j \geq 0} \frac{H^{(j)}(R)}{j!} \delta^j = \sum_{n \geq 0} b_n (R + \delta)^n < \infty.$$

The original power series would converge beyond R , a contradiction. This is the usual Pringsheim argument, and it proves that H is entire.

Now $0 \leq b_n t^n \leq H(t)$ for every positive real t . A polynomial upper bound $H(t) = O(t^M)$ forces $b_n = 0$ for every $n > M$. Thus H , and hence g , is a polynomial, the final contradiction. Eventual nonnegativity and eventual nonpositivity have both been excluded. $\square$

Theorem 8.6 (Infinite diagonal oscillation). *For every $A > 1$, each of the sets*

$$\{n \geq 1 : v_n(A) > 0\}, \quad \{n \geq 1 : v_n(A) < 0\}$$

is infinite. Consequently, for every lattice spacing $h > 0$, the smallest zero branch of $\mathcal{F}_{n,n-1}^{\rho,h}$ initially moves in each direction for infinitely many Laurent indices $n \geq 2$.

Proof. Set $L = \log A > 0$ and consider

$$h_A(u) = \frac{e^u - A}{u}, \quad u \geq L.$$

It has $h_A(L) = 0$ and tends to infinity. Its derivative is

$$h'_A(u) = \frac{e^u(u-1) + A}{u^2} > 0.$$

For the strict inequality, the numerator equals $AL > 0$ at $u = L$, and its derivative is $ue^u > 0$ for $u > 0$. Thus h_A has a unique increasing real inverse on $[0, \infty)$. The derivative never vanishes, so the complex implicit function theorem continues this inverse holomorphically near every positive real point. It is the continuation of u_A from (8.9).

The inverse is unbounded but has logarithmic growth. To see the latter directly, take $U = 2\log(t + A + 2)$. For all sufficiently large t ,

$$e^U = (t + A + 2)^2 > tU + A,$$

so $h_A(U) > t$ and monotonicity implies $u_A(t) < 2\log(t + A + 2)$. An unbounded function of this growth cannot be a polynomial. Lemma 8.5 applies to u_A and proves infinitely many Taylor coefficients of each sign. Equation (8.10) negates all its nonconstant coefficients. This proves the result for $v_n(A)$; omission of the single index $n = 1$ does not change infinitude. $\square$

Corollary 8.7 (A restriction on uniform monotonicity thresholds). *For the unit lattice, no integer N has the property that every smallest diagonal branch $a_n^{(1)}(\rho)$, $n \geq N$, is nonincreasing near $\rho = 0$. If K_n is any threshold guaranteeing decreasing motion of every zero branch throughout the Lerch interval for every $k \geq K_n$, then $K_n > n - 1$ for infinitely many n .*

Proof. For infinitely many n , Theorem 8.6 gives $(a_n^{(1)})'(0) > 0$. Analyticity makes the branch strictly increasing on a sufficiently small positive interval, contradicting each of the proposed conclusions when $k = n - 1$ is included. $\square$

This conclusion respects the quantifiers of the earlier fixed-index theorem. It does not assert that its threshold fails to exist, or that the diagonal branch must continue simply throughout $0 \leq \rho \leq 1$. The local interval in Lemma 8.1 can depend on n .

8.4 A short exact example and the verification artifacts

For the unit lattice put $L = \log 2$. The first relevant velocities are

$$\begin{aligned} v_2(2) &= \frac{L^2 - 2L}{8}, \\ v_3(2) &= \frac{-2L^3 + 9L^2 - 6L}{48}, \\ v_4(2) &= \frac{3L^4 - 22L^3 + 36L^2 - 12L}{192}. \end{aligned}$$

The fourth-index example is positive:

$$0.01221096503 < v_4(2) < 0.01221096504. \quad (8.13)$$

For a reproducible exact certificate, use

$$L_M = 2 \sum_{j=0}^{M-1} \frac{1}{(2j+1)3^{2j+1}}, \quad 0 < \log 2 - L_M < \frac{2}{(2M+1)3^{2M+1}(1-1/9)}.$$

This is the positive series for $2 \operatorname{arctanh}(1/3)$ with a geometric tail majorant. At $M = 20$, rational interval Horner evaluation of the displayed polynomial gives (8.13). Equivalently,

$$P_{4,3}(x) = x^4 + \frac{22}{3}x^3 + 12x^2 + 4x, \quad v_4(2) = \frac{P_{4,3}(-\log 2)}{64}.$$

The exact verifier compares the Stirling expression with the quadratic recurrence over $\mathbb{Q}[L]$ through $n = 40$. It also checks the inverse generating equation through degree forty at a formal rational value of L , and encloses the actual velocities at $L = \log 2$ by rational arithmetic. The signs for $2 \leq n \leq 12$ are

n	2	3	4	5	6	7	8	9	10	11	12
$\operatorname{sgn} v_n(2)$	−	−	+	+	+	−	−	+	+	+	−

These finite certificates check the implementation and the short example. The infinite sign conclusion rests on the analytic proof above.

8.5 Questions left by the inverse representation

The exact generating function permits a quantitative refinement, but its complex branches must be handled carefully. A finite critical point of the inverse equation satisfies

$$(1 - u)e^u = A, \quad t = e^u. \quad (8.14)$$

Different Lambert- W sheets give different critical values, some tending to zero on other sheets. Therefore one cannot read off the radius of the germ $u_A(t)$ by minimizing over all formal critical values. The oscillation theorem does not require such a dominance assertion.

Concrete further questions are to identify the accessible dominant singularities and obtain asymptotic formulas for $v_n(A)$; to bound gaps between indices with opposite velocity signs; to determine a uniform neighborhood in which the first velocity controls the true branch; and to extend the inverse-series method to $n = k + r$, $r > 1$, where the central elementary root has multiplicity r and its unfolding is generally described by fractional powers. These questions connect the central branch treated here with the existing positive-Bessel-zero regime.

9 A quantitative conjecture for the diagonal velocities

The inverse-function identity for the diagonal branch suggests a much sharper description of its oscillations. This section separates the exact local calculation from the global analytic-continuation problem that is still open. We restrict to the original lattice spacing $h = 1$, so $A = 2$. Write

$$V(t) = \sum_{n \geq 1} v_n(2)t^n = \log 2 - u(t), \quad e^{u(t)} = 2 + tu(t), \quad u(0) = \log 2. \quad (9.1)$$

Thus the coefficient formula, independently of any conjecture below, is

$$v_n(2) = 2^{-n} \sum_{j=1}^n \left[\begin{matrix} n \\ n-j+1 \end{matrix} \right] \frac{(-\log 2)^j}{j!}, \quad (9.2)$$

where $\left[\begin{matrix} n \\ k \end{matrix} \right]$ denotes an unsigned Stirling number of the first kind.

9.1 The candidate critical pair and its local expansion

A finite critical point of the inverse relation must satisfy

$$e^{u_*} = t_*, \quad (1 - u_*)e^{u_*} = 2. \quad (9.3)$$

We specify one conjugate pair without relying on a branch convention for Lambert's W function. Let $\theta \in (0, \pi/2)$ be the solution of

$$1 - \theta \cot \theta + \log \left(\frac{\theta}{\sin \theta} \right) = \log 2, \quad (9.4)$$

and put

$$a = 1 - \theta \cot \theta, \quad u_* = a + i\theta, \quad t_* = e^{u_*}, \quad R = |t_*| = e^a. \quad (9.5)$$

These definitions are exact. The left side of (9.4) increases strictly from 0 to $1 + \log(\pi/2)$, since its derivative is

$$\theta \csc^2 \theta + \frac{1}{\theta} - 2 \cot \theta = \frac{(\theta - \sin \theta \cos \theta)^2 + \sin^4 \theta}{\theta \sin^2 \theta} > 0.$$

Consequently (9.4) has exactly one solution in the stated interval. Direct substitution proves (9.3). Equivalently, $u_* = 1 + W_0(-2/e + i0)$, using the upper boundary value of the principal branch of Lambert's function [17, §4.13].

The following decimal values are numerical evaluations of the exact definitions, not replacements for them:

$$\begin{aligned} u_* &\approx 0.469353635658274 + 1.132672497604881i, \\ t_* &\approx 0.678344933205449 + 1.447937273036010i, \\ R &\approx 1.598960348180173. \end{aligned}$$

Proposition 9.1 (Exact local square-root calculation). *Let $z = 1 - t/t_*$. The two local inverse branches meeting at (t_*, u_*) are convergent power series in $z^{1/2}$. For the choice*

$$\chi = \sqrt{-2u_*}, \quad \Re\chi > 0, \quad \Im\chi < 0, \quad (9.6)$$

one of these branches has expansion

$$u(t) = u_* + \chi z^{1/2} + bz + d_3 z^{3/2} + b_2 z^2 + O(z^{5/2}), \quad (9.7)$$

where b_2 is a constant and

$$b = \frac{u_*}{3} - 1, \quad d_3 = \chi \left(\frac{1}{2} - \frac{u_*}{18} - \frac{1}{4u_*} \right). \quad (9.8)$$

The other branch changes the signs of all odd powers of $z^{1/2}$. This statement does not identify either local branch with the germ in (9.1).

Proof. Set $d = u - u_*$. Dividing the implicit equation by t_* and using (9.3) gives the exact local equation

$$e^d - 1 - d + z(u_* + d) = 0. \quad (9.9)$$

Equivalently, $z = -(e^d - 1 - d)/(u_* + d) = -d^2/(2u_*) + O(d^3)$. Since $u_* \neq 0$, the local analytic inverse is a convergent power series in $\sqrt{z}$, with its two possible signs. Substitution of $d = \chi z^{1/2} + bz + d_3 z^{3/2} + \dots$ into (9.9) gives successively

$$\frac{\chi^2}{2} + u_* = 0, \quad \chi b + \frac{\chi^3}{6} + \chi = 0,$$

and

$$\chi d_3 + \frac{b^2}{2} + \frac{\chi^2 b}{2} + \frac{\chi^4}{24} + b = 0.$$

These identities give (9.6) and (9.8). □

Numerically,

$$\begin{aligned} \chi &\approx 0.869892622200723 - 1.302083117729355i, \\ d_3 &\approx 0.507703114841215 - 0.406328525288279i. \end{aligned}$$

There are infinitely many other formal critical values coming from the other solutions of (9.3). They have the form $t_k = -2/W_k(-2/e)$, so the values corresponding to sufficiently remote Lambert branches tend to zero [17, §4.13]. They cannot all be singularities of the germ (9.1), which is holomorphic near zero. Their relevance depends on which sheet of the inverse is reached. Listing critical values and minimizing their moduli therefore does not settle the radius of convergence of V .

9.2 An explicit dominance conjecture and its consequence

Conjecture 9.2 (Accessible dominant conjugate pair for $A = 2$). For some $\epsilon > 0$, the germ u in (9.1) has a single-valued holomorphic continuation to the slit disk

$$\Omega_\epsilon = \{t : |t| < R + \epsilon\} \setminus (\{st_* : 1 \leq s < 1 + \epsilon/R\} \cup \{\overline{st_*} : 1 \leq s < 1 + \epsilon/R\}). \quad (9.10)$$

Its limiting values at the two slit endpoints are respectively u_* and $\overline{u_*}$. Near the upper endpoint it is the branch (9.7) with χ chosen in (9.6), where $z^{1/2} > 0$ along $t = st_*$ for $s < 1$ sufficiently close to 1. At the lower endpoint its expansion is the complex conjugate.

The conjecture includes three substantive claims: accessibility of the specified critical points from the germ at zero, the identification of the square-root sign, and absence of other singularities on or inside the circle of radius R . None is inferred here from the formal critical equations alone.

Theorem 9.3 (Conditional two-term coefficient asymptotic). *Assume Conjecture 9.2. Define*

$$\chi_1 = \frac{3\chi}{8} - \frac{3d_3}{2} = \chi \left(-\frac{3}{8} + \frac{u_*}{12} + \frac{3}{8u_*} \right). \quad (9.11)$$

Then the radius of convergence of V is R , and

$$v_n(2) = \frac{R^{-n}}{\sqrt{\pi} n^{3/2}} \Re \left[e^{-in\theta} \left(\chi + \frac{\chi_1}{n} \right) \right] + O(R^{-n} n^{-7/2}). \quad (9.12)$$

In particular, if $\phi = -\arg \chi$, then

$$v_n(2) = \frac{|\chi|}{\sqrt{\pi} R^n n^{3/2}} \cos(n\theta + \phi) + O(R^{-n} n^{-5/2}). \quad (9.13)$$

The errors are additive; no division by the oscillating cosine is intended.

Proof. The conjectured continuation and the nonzero square-root coefficient give radius exactly R . At t_* the nonintegral terms in $V = \log 2 - u$ are $-\chi z^{1/2} - d_3 z^{3/2}$, with conjugate terms at $\overline{t_*}$. The exact binomial-coefficient identity and its gamma ratio expansion give, for either nonintegral exponent used here,

$$[t^n](1 - t/t_*)^\alpha = \frac{t_*^{-n} n^{-\alpha-1}}{\Gamma(-\alpha)} \left(1 + \frac{\alpha(\alpha+1)}{2n} + O(n^{-2}) \right).$$

Using $\Gamma(-1/2) = -2\sqrt{\pi}$ and $\Gamma(-3/2) = 4\sqrt{\pi}/3$, the upper singularity contributes

$$\frac{t_*^{-n}}{2\sqrt{\pi} n^{3/2}} \left(\chi + \frac{3\chi/8 - 3d_3/2}{n} \right) + O(R^{-n} n^{-7/2}).$$

Adding the conjugate contribution proves (9.12). The slit disk contains the two dented neighborhoods needed for singularity transfer, and the convergent Puiseux expansion supplies the remainder estimate. The integral powers are locally holomorphic and do not contribute to these algebraic coefficient terms. This is the finite-singularity form of the transfer method of Flajolet and Odlyzko [16]. Taking the real part of $\chi e^{-in\theta}$ gives (9.13) with the stated sign. $\square$

The numerical amplitude and phase are

$$\frac{|\chi|}{\sqrt{\pi}} \approx 0.883480827960335, \quad \theta \approx 1.132672497604881, \quad \phi \approx 0.981817528486668,$$

and $\chi_1 \approx -0.435344938936552 + 0.121211618783910i$. Thus the conjecture predicts an exponentially decaying, algebraically modulated oscillation. The unconditional proof of infinitely many positive and negative velocities does not require this conjecture.

9.3 Exact coefficient certificates and numerical diagnostics

The accompanying script code/diagonal_quantitative.py evaluates (9.2) for every integer $50 \leq n \leq 200$ using exact integer interval arithmetic. It encloses $\log 2$ by a rational truncation of

$$\log 2 = 2 \sum_{j=0}^{\infty} \frac{1}{(2j+1)3^{2j+1}},$$

with a geometric upper bound for the tail, and evaluates the Stirling polynomials by Horner's rule with outward rounding on a grid of size 10^{-400} . Every resulting interval excludes zero; its relative width is less than 10^{-330} by an exact integer comparison. All interval endpoints are included in data/diagonal_quantitative.json. An independently implemented coefficient recurrence agrees numerically with their midpoints to over 180 relative decimal places.

For the asymptotic comparison, put

$$Z_n = \sqrt{\pi} R^n n^{3/2} v_n(2), \quad Z_n^{(0)} = \Re(\chi e^{-in\theta}), \quad Z_n^{(1)} = \Re((\chi + \chi_1/n) e^{-in\theta}).$$

The critical parameters and the model errors below were evaluated with 280-digit arithmetic; those model calculations are numerical, in contrast with the exact coefficient intervals. Normalized absolute errors avoid spurious large relative errors near zeros of the cosine.

n	$v_n(2)$, approximately	$ Z_n - Z_n^{(0)} $	$ Z_n - Z_n^{(1)} $
50	$7.671927522997 \cdot 10^{-14}$	$8.4090050 \cdot 10^{-3}$	$6.078260 \cdot 10^{-5}$
75	$-3.097444735256 \cdot 10^{-19}$	$5.5287188 \cdot 10^{-3}$	$2.341243 \cdot 10^{-5}$
100	$1.476168203925 \cdot 10^{-24}$	$4.0746493 \cdot 10^{-3}$	$1.114101 \cdot 10^{-5}$
125	$-7.660459633287 \cdot 10^{-30}$	$3.1942636 \cdot 10^{-3}$	$5.825053 \cdot 10^{-6}$
150	$4.168465378193 \cdot 10^{-35}$	$2.6016760 \cdot 10^{-3}$	$3.133644 \cdot 10^{-6}$
175	$-2.325940260211 \cdot 10^{-40}$	$2.1739529 \cdot 10^{-3}$	$1.629365 \cdot 10^{-6}$
200	$1.309142927750 \cdot 10^{-45}$	$1.8494967 \cdot 10^{-3}$	$7.305141 \cdot 10^{-7}$

Table 2: Diagnostic errors for the conjectured velocity asymptotic; the coefficient signs themselves have exact interval certificates.

Across all 151 consecutive indices, the observed maxima are

$$\begin{aligned} \max_{50 \leq n \leq 200} n |Z_n - Z_n^{(0)}| &\approx 0.4498059481, \\ \max_{50 \leq n \leq 200} n^2 |Z_n - Z_n^{(1)}| &\approx 0.4839906969. \end{aligned}$$

The leading model's computed sign agrees with the certified sign of $v_n(2)$ at every tested index. The smallest tested value of $|Z_n^{(0)}|$ is approximately 0.0138238350 at $n = 56$; this finite sample does not explore arbitrarily close approaches to a cosine zero.

As a separate diagnostic of the local sign, the script follows the root of (9.1) numerically along $t = st_*$, starting at $u(0) = \log 2$ and using successive warm Newton starts. At the last tested point, $1 - s \approx 3.76 \cdot 10^{-38}$, the estimate $(u(st_*) - u_*)/\sqrt{1-s}$ differs from χ by approximately $1.8 \cdot 10^{-19}$, consistent with the next term of (9.7). This computation was not initialized by imposing the chosen local square-root sign.

These data support the proposed pair, local sign, and two coefficient terms. They do not certify continuation of a branch, prevent a numerical root solver from changing sheets, or exclude an additional singularity of smaller or equal modulus. The main remaining analytic problem is to prove or refute Conjecture 9.2 on the inverse surface. Even if it holds, questions about the density of positive velocities require control of the rotation angle and of indices close to cosine zeros; sign agreement on a finite range settles neither issue.

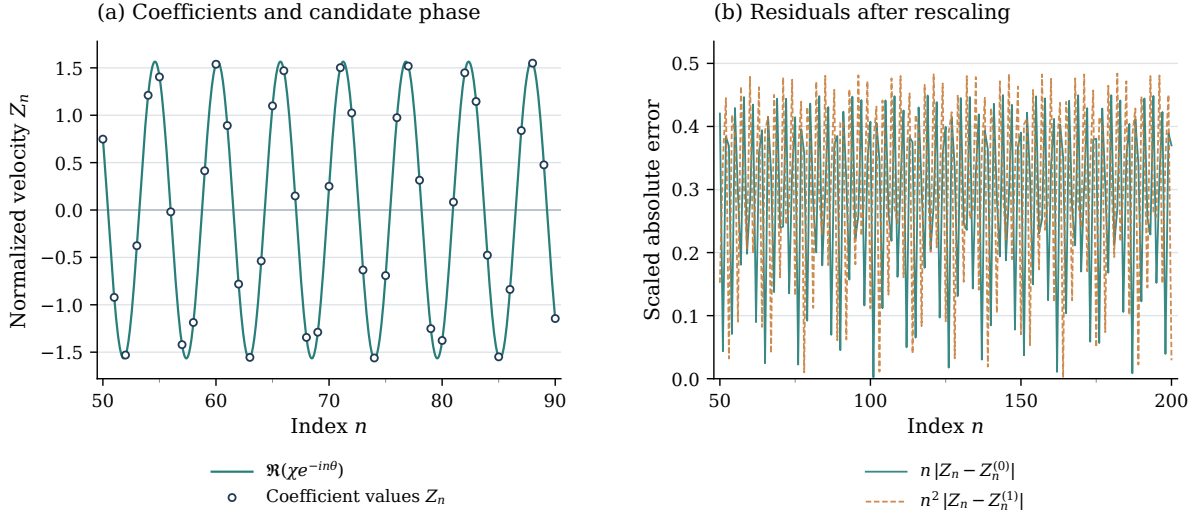


Figure 1: Diagonal velocities and the proposed dominant conjugate pair. Left: normalized coefficient values and the candidate oscillatory phase. Right: leading and corrected normalized errors after multiplication by n and n^2 . The coefficients have exact interval certificates; the normalization and asymptotic comparison are numerical diagnostics of Conjecture 9.2, not a proof of accessibility or dominance.

10 The optimal exterior differential barrier

The incoming report *Lerch Global Phase* proves an exterior one-crossing inequality with the fixed coefficient e^{-n} and explicitly asks for the optimum over all positive coefficients in its further-research question 2. We solve that optimization problem. The sharp coefficient is controlled by a cubic algebraic number; its large-index limit involves the golden ratio. A distinction between a strict pointwise inequality and a strict inequality for positive geometric sums is essential at the optimum.

For integers $n \geq 2$ define

$$g_n(x) = \frac{(\log x)^{n-1}(n - \log x)}{x^2}, \quad U_n(a, \rho) = \sum_{m=0}^{\infty} \rho^m g_n(a + m). \quad (10.1)$$

Here $a > 0$ and $0 \leq \rho \leq 1$. The series and each fixed number of its a derivatives converge absolutely and locally uniformly, uniformly in ρ , since the m th summand after j derivatives is $O(m^{-2-j}(1 + \log^n m))$. For $\rho < 1$, U_n is the first a derivative of the order- n Lerch coefficient $\sum_{m \geq 0} \rho^m \log^n(a + m)/(a + m)$; at $\rho = 1$ it is $\gamma'_n(a)$ in the usual Stieltjes Laurent convention.

Call a pair (B, λ) *uniformly admissible* if $B > 0$, $\lambda > 0$, and

$$\partial_a U_n(a, \rho) + \lambda U_n(a, \rho) < 0 \quad (a \geq B, 0 < \rho \leq 1). \quad (10.2)$$

The uniformity down to arbitrarily small positive ρ makes this a precise optimization problem. A smaller cutoff at one fixed deformation parameter is a different problem.

Put $\varphi = (1 + \sqrt{5})/2$. Let r_n be the unique positive root of

$$r^3 + nr^2 - nr - n = 0, \quad (10.3)$$

and set

$$\tau_n = (2 - r_n)e^{-r_n}, \quad \lambda_n = e^{-n}\tau_n, \quad \Delta_n = \sqrt{n^2 + 8n}, \quad \alpha_n = \frac{3n - \Delta_n}{4}. \quad (10.4)$$

For $0 < \tau \leq 1$, write

$$R_{n,\tau}(y) = 2y^2 - 3ny + n(n-1) + \tau e^{y-n}y(n-y). \quad (10.5)$$

Theorem 10.1 (Optimal coefficient and optimal cutoff). *For every integer $n \geq 2$ the following statements hold.*

1. *The cubic root satisfies $1 < r_n < \varphi$. There is a unique $\eta_n \in (\alpha_n, (n-1)/2)$ such that $R_{n,\tau_n}(\eta_n) = 0$. Put $B_n = e^{\eta_n}$.*
2. *The pointwise inequality is*

$$g'_n(x) + \lambda_n g_n(x) \leq 0 \quad (x \geq B_n). \quad (10.6)$$

Equality occurs at exactly two points,

$$x = B_n \quad \text{and} \quad x = e^{n+r_n}. \quad (10.7)$$

3. *The pair (B_n, λ_n) is uniformly admissible. Every uniformly admissible pair (B, λ) has $B \geq B_n$, and if $B = B_n$ then $\lambda = \lambda_n$.*
4. *Consequently $e^{\lambda_n a} U_n(a, \rho)$ is strictly decreasing on $[B_n, \infty)$ for every $0 < \rho \leq 1$. There is at most one zero there; any such zero is simple and crosses from positive to negative.*

We give a complete proof, including the boundary qualification.

Lemma 10.2 (The negative-tail constraint). *For $r > 0$ put*

$$T_n(r) = 2 - \frac{n-1}{n+r} - \frac{1}{r}.$$

The function $e^{-r} T_n(r)$ has a unique maximum, at $r = r_n$, and

$$\max_{r>0} e^{-r} T_n(r) = (2 - r_n) e^{-r_n} = \tau_n. \quad (10.8)$$

Thus λ_n is the smallest coefficient that can make $g'_n + \lambda g_n \leq 0$ on the whole interval (e^n, ∞) .

Proof. For $x = e^{n+r} > e^n$ one has $g_n(x) < 0$ and

$$-\frac{g'_n(x)}{g_n(x)} = e^{-n-r} T_n(r).$$

Dividing the desired inequality by the negative quantity $g_n(x)$ shows that it is equivalent to $\lambda \geq e^{-n-r} T_n(r)$ for every $r > 0$.

Now

$$T'_n(r) = \frac{n-1}{(n+r)^2} + \frac{1}{r^2} > 0, \quad T''_n(r) = -\frac{2(n-1)}{(n+r)^3} - \frac{2}{r^3} < 0.$$

Therefore $T'_n - T_n$ is strictly decreasing, tends to $+\infty$ at zero, and tends to -2 at infinity. It has precisely one zero, which is the unique maximum of $e^{-r} T_n(r)$. The factorization

$$(T_n - T'_n) r^2 (n+r)^2 = (n+2r)(r^3 + nr^2 - nr - n) \quad (10.9)$$

identifies that point with the positive root of (10.3). The cubic has exactly one positive root by its coefficient sign variation, or by the preceding uniqueness argument. Its values at 1 and φ are $1 - n < 0$ and $\varphi^3 > 0$, respectively, giving $1 < r_n < \varphi$.

Rearranging the cubic gives $n = r_n^3 / (1 + r_n - r_n^2)$. Substitution in $T_n(r_n)$ simplifies it to $2 - r_n$, proving (10.8). The maximum is positive; the tail inequality is strict except at its unique maximizing point. $\square$

Lemma 10.3 (The positive-side cutoff). *For every $n \geq 2$ and every $0 < \tau \leq 1$, the function $R_{n,\tau}$ has exactly one root $\eta_n(\tau)$ in $(0, n)$. It satisfies*

$$\alpha_n < \eta_n(\tau) < \frac{n-1}{2}, \quad R_{n,\tau}(y) > 0 \quad (0 < y < \eta_n(\tau)), \quad R_{n,\tau}(y) < 0 \quad (\eta_n(\tau) < y \leq n).$$

The root is simple, and $\eta_n(\tau)$ is strictly increasing in τ .

Proof. For $y = n - t$, $0 \leq t \leq (n+1)/2$, use $\tau e^{-t} \leq (1+t)^{-1}$ and $(n-t)t \geq 0$. This gives

$$\begin{aligned} (1+t)R_{n,\tau}(n-t) &\leq 2t^3 + (1-n)t^2 - nt - n \\ &\leq 2t^2 - nt - n < 0. \end{aligned}$$

For the second inequality use $2t \leq n+1$. The final quadratic is convex, with negative values $-n$ and $(1-n)/2$ at the two endpoints, so it is negative throughout. This proves negativity for $(n-1)/2 \leq y \leq n$.

On $0 \leq y \leq (n-1)/2$,

$$R''_{n,\tau}(y) = 4 + \tau e^{y-n} \{-y^2 + (n-4)y + 2n - 2\} > 4.$$

The bracket is a concave quadratic whose endpoint values are $2(n-1)$ and $(n^2-1)/4$, both positive. Since $R_{n,\tau}(0) > 0$ and $R_{n,\tau}((n-1)/2) < 0$, strict convexity gives one and only one crossing on this interval; a second crossing would force the function to be nonnegative at its right endpoint. The derivative at the crossing is strictly negative. The polynomial part vanishes at α_n , so $R_{n,\tau}(\alpha_n) > 0$, proving the lower bound. Finally

$$\frac{\partial \eta_n(\tau)}{\partial \tau} = -\frac{e^{\eta_n(\tau)-n} \eta_n(\tau)(n-\eta_n(\tau))}{\partial_y R_{n,\tau}(\eta_n(\tau))} > 0.$$

□

Proof of Theorem 10.1. For $x = e^y > 1$, differentiation gives

$$g'_n(x) + \tau e^{-n} g_n(x) = x^{-3} y^{n-2} R_{n,\tau}(y). \quad (10.10)$$

The prefactor is positive. Because $1 < r_n < \varphi < 2$, $0 < \tau_n < 1$. Lemma 10.3 treats the region below e^n , Lemma 10.2 treats the region above it, and at $x = e^n$ one has $g'_n < 0$. This proves the nonpositive inequality and the exact equality set.

For $a \geq B_n$, every summand in

$$(\partial_a + \lambda_n)U_n(a, \rho) = \sum_{m=0}^{\infty} \rho^m [g'_n(a+m) + \lambda_n g_n(a+m)]$$

is nonpositive. Only two possible arguments can give equality. When $\rho > 0$, at least one of the three summands $m = 0, 1, 2$ is strictly negative, so the sum is strictly negative. Absolute convergence justifies the differentiation and summation, also at $\rho = 1$.

To prove optimality, suppose first that $\lambda < \lambda_n$. At $x_* = e^{n+r_n}$, where $g_n(x_*) < 0$, one has $g'_n(x_*) + \lambda g_n(x_*) > 0$. Since $(\partial_a + \lambda)U_n(a, \rho)$ tends to this pointwise expression as $\rho \downarrow 0$, every cutoff $B \leq x_*$ fails for small positive ρ . In particular, such a coefficient cannot beat B_n . If $\lambda \geq \lambda_n$ and $B < B_n$, choose $\max(B, 1) < a < B_n$. Then $g_n(a) > 0$ and

$$g'_n(a) + \lambda g_n(a) \geq g'_n(a) + \lambda_n g_n(a) > 0,$$

again contradicting admissibility at sufficiently small positive ρ . This proves $B \geq B_n$. At $B = B_n$, any $\lambda > \lambda_n$ makes the pointwise expression strictly positive at $a = B_n$, so only λ_n is possible. The integrating factor and the sign of the derivative at any zero prove the last assertion. □

Why the pointwise strict optimum is not attained. If the requirement is instead $g'_n(x) + \lambda g_n(x) < 0$ for every $x > B$, the infimum of possible cutoffs is still B_n , but no coefficient attains it. At λ_n the outer point $e^{n+r_n} > B_n$ has equality. Coefficients $\lambda > \lambda_n$, sufficiently close to λ_n , have strict tail inequality and cutoff $e^{\eta_n(e^n \lambda)}$ decreasing to B_n as $\lambda \downarrow \lambda_n$. The strict summed theorem therefore must not be copied verbatim into a statement about every positive measure or about the elementary endpoint $\rho = 0$.

Positive measures. Let ν be a nonzero positive measure on $[0, \infty)$ satisfying

$$\int_{[0, \infty)} \frac{1 + \log^n(2+t)}{(1+t)^2} d\nu(t) < \infty.$$

This condition supplies locally uniform domination of the integral and its derivative. The function $H_n(a) = \int g_n(a+t) d\nu(t)$ satisfies $H'_n + \lambda_n H_n \leq 0$ for $a \geq B_n$. It is strictly negative unless the measure is supported on the at most two points for which $a+t$ belongs to (10.7). In particular, non-atomic measures and measures with at least three support points give the strict inequality. This is the exact measure-theoretic qualification forced by the optimal tangency.

10.1 The golden-ratio limit and convergent coefficient formulas

Theorem 10.4 (Monotone optimizer and its large-index expansion). *The sequence r_n strictly increases to φ , whereas $\tau_n = e^n \lambda_n$ strictly decreases to $\tau_\infty = (2 - \varphi)e^{-\varphi}$. As $n \rightarrow \infty$,*

$$r_n = \varphi - \left(1 + \frac{2\sqrt{5}}{5}\right) \frac{1}{n} + \left(\frac{5}{2} + \frac{57\sqrt{5}}{50}\right) \frac{1}{n^2} + O(n^{-3}), \quad (10.11)$$

$$\lambda_n = e^{-n} \tau_\infty \left[1 + \frac{7 + 3\sqrt{5}}{2n} - \frac{35 + 16\sqrt{5}}{10n^2} + O(n^{-3})\right]. \quad (10.12)$$

Writing $\epsilon = 1/n$, the complete expansion of r_n is a convergent power series for $|\epsilon| < 5/27$. Its coefficients are

$$r(\epsilon) = \varphi + \sum_{j \geq 1} \frac{(-\epsilon)^j}{j} [z^{j-1}] \left(\frac{(\varphi + z)^3}{\sqrt{5} + z} \right)^j. \quad (10.13)$$

In particular this is a convergent formula for every integer $n \geq 6$.

Proof. The identity $n = r^3/(1+r-r^2)$ has strictly positive derivative on $1 < r < \varphi$:

$$\frac{d}{dr} \frac{r^3}{1+r-r^2} = \frac{r^2(3+2r-r^2)}{(1+r-r^2)^2} > 0.$$

It tends to infinity at φ , proving the claims about r_n . Since $d[(2-r)e^{-r}]/dr = (r-3)e^{-r} < 0$ in this interval, the claims about τ_n follow.

The implicit equation $\epsilon r^3 + r^2 - r - 1 = 0$ has derivative $\sqrt{5}$ in r at $(\epsilon, r) = (0, \varphi)$, so the analytic implicit-function theorem applies. Substitution of a quadratic series gives (10.11), and substitution into $(2-r)e^{-r}$ gives (10.12). With $r = \varphi + z$ the equation becomes

$$z = -\epsilon \frac{(\varphi + z)^3}{\sqrt{5} + z}.$$

Lagrange inversion gives (10.13).

For completeness the convergence radius can be identified exactly. The discriminant in r is

$$\text{disc}_r(\epsilon r^3 + r^2 - r - 1) = (1 - \epsilon)(27\epsilon + 5).$$

The branch selected at zero is holomorphic throughout $|\epsilon| < 5/27$: away from zero the discriminant is nonzero there, and the selected branch has a removable, already analytic continuation at zero. Along the real interval $-5/27 < \epsilon \leq 0$ this branch increases from φ to 3. At $(\epsilon, r) = (-5/27, 3)$, the first r derivative vanishes while $\partial_\epsilon = 27$ and $\partial_r^2 = -4/3$ are nonzero. It is therefore a genuine square-root branch point. Hence the Taylor radius is exactly $5/27$. $\square$

10.2 The exponentially small displacement of the cutoff

The optimizer has an ordinary convergent expansion in $1/n$. The cutoff also contains an exponentially small displacement in logarithmic coordinates, which should be kept separate from that algebraic expansion. Put

$$E_n = e^{\alpha_n - n}, \quad A_n = \alpha_n(n - \alpha_n), \quad C_n = n - 2\alpha_n.$$

Theorem 10.5 (Cutoff expansion and an all-orders formula). *As $n \rightarrow \infty$,*

$$\begin{aligned} \eta_n = \alpha_n + \frac{\tau_n A_n}{\Delta_n} E_n \\ + \tau_n^2 \left[\frac{A_n(A_n + C_n)}{\Delta_n^2} + \frac{2A_n^2}{\Delta_n^3} \right] E_n^2 + O(n^3 E_n^3). \end{aligned} \quad (10.14)$$

Every further power of E_n has an explicit coefficient. More precisely, for all sufficiently large n the solution $\delta = \eta_n - \alpha_n$ is obtained, at $E = E_n$, from the locally convergent series

$$\delta(E) = \sum_{j \geq 1} \frac{E^j}{j} [z^{j-1}] \left[\frac{\tau_n e^z (\alpha_n + z)(n - \alpha_n - z)}{\Delta_n - 2z} \right]^j. \quad (10.15)$$

For each fixed truncation order J , its error at E_n is $O_J(n^{J+1} E_n^{J+1})$. In the original a coordinate this implies

$$B_n = e^{\alpha_n} + \frac{\tau_\infty}{4e^2} \left(n + \frac{7 + 3\sqrt{5}}{2} \right) + O(n^{-1}). \quad (10.16)$$

The leading additive displacement above the first stationary point of g_n has coefficient

$$\frac{\tau_\infty}{4e^2} = 0.0025625510827 \dots$$

Proof. Substitute $\eta_n = \alpha_n + \delta$ in its defining equation and use the exact quadratic Taylor expansion of the polynomial part:

$$-\Delta_n \delta + 2\delta^2 + \tau_n E_n e^\delta (\alpha_n + \delta)(n - \alpha_n - \delta) = 0.$$

Equivalently, $\delta = E_n f_n(\delta)$ with

$$f_n(z) = \frac{\tau_n e^z (\alpha_n + z)(n - \alpha_n - z)}{\Delta_n - 2z}.$$

This proves (10.15) by Lagrange inversion. Its first two coefficients are $f_n(0)$ and $f_n(0)f'_n(0)$, giving (10.14).

The remainder statement can be justified uniformly in n . On a fixed small complex disk in z , f_n and its derivative are $O(n)$, while $\Delta_n \asymp n$. For $|E| < c/n$ with a sufficiently small absolute $c > 0$, the map $z \mapsto E f_n(z)$ is a contraction of a smaller fixed disk. The analytic solution is uniformly bounded there. Cauchy estimates therefore bound its j th coefficient by $O(C^j n^j)$. Since $E_n \asymp e^{-n/2}$, every fixed-order tail has the asserted bound for sufficiently large n .

Finally exponentiate the first sector. Its omitted terms become $O(n^2 e^{-n/2})$ after multiplication by e^{α_n} . Direct expansion gives

$$\frac{A_n}{\Delta_n} e^{2\alpha_n - n} = e^{-2} \left(\frac{n}{4} - \frac{1}{n} + O(n^{-2}) \right).$$

Combining this with (10.12) yields (10.16). No relative approximation to $\eta_n - \alpha_n$ is being inferred from an uncontrolled algebraic remainder in α_n . $\square$

Limits of the conclusion. The optimum is over constant first-order operators, required uniformly for all positive geometric weights down to $\rho = 0$. It does not optimize variable-coefficient operators, inequalities only at the zeros, or cutoffs at one fixed ρ . Those extensions remain separate research questions. In particular, improving an exterior barrier does not on its own classify all interior zeros or prove a global monotonicity theorem for individual zero branches.

11 Reproducible computations and numerical illustrations

11.1 What the checks establish

The package separates three different forms of evidence. The general theorems are proved in the preceding sections. Exact finite checks verify rational identities or enclose particular numbers using a proved remainder bound. Floating-point computations check conventions and illustrate asymptotic behavior. The last category is not used to certify equality, a zero count, or a decimal enclosure.

The harmonic verifier uses two independent exact constructions through order thirty. One differentiates the kernel in the form

$$K^{(j)}(t) = \frac{\sqrt{y}}{(1+y)^{j+1}} \{P_j(y) \log(1+y) + Q_j(y)\}, \quad y = e^{-2t}.$$

The other constructs $\sec z$ and $\sec z \log \sec z$ by rational formal power-series operations. The resulting rational and $\log 2$ coefficients agree at all thirty-one nonpositive integer arguments. The same program checks the polynomial in the proof of $K'' > 0$, the rational constant $181/27$, and the right-half-plane tail bound.

The diagonal verifier compares the Stirling expression with the independently derived quadratic recurrence in $\mathbb{Q}[L]$ through $n = 40$. It also checks the inverse generating equation through degree forty after a rational formal specialization. A positive-series enclosure of $\log 2$, followed by rational interval Horner evaluation, certifies the velocity signs for $2 \leq n \leq 40$. The separate quantitative experiment extends exact sign enclosures to $50 \leq n \leq 200$ and compares them with the conjectured singularity asymptotic. Its analytic continuation and asymptotic comparisons remain numerical diagnostics.

The barrier verifier checks ten exact symbolic identities, including the critical-point factorization, the cubic discriminant, and the displayed inverse-index coefficients. For $n = 2, 3, 4, 5, 10, 25, 100$ it also isolates the optimizer and cutoff by exact rational interval arithmetic. The exponential evaluator reduces its argument to $[0, 1]$, uses a Taylor polynomial with the explicit tail bound $2/(M+1)!$, and then applies outward-rounded inversion and repeated squaring. Thus the recorded decimal endpoints are consequences of rational comparisons, rather than the output of a floating-point root finder.

11.2 Harmonic zeros and the two asymptotic scales

Table 3 lists diagnostic roots from the positive Mellin-moment ratio. The first three were also substituted into independent differentiated-kernel Mellin integrals. Those checks do not share the digamma representation. At forty-five working decimal digits, the three independent absolute residuals are approximately 2.1×10^{-43} , 1.1×10^{-31} , and 1.2×10^{-25} . Working precision is not a guarantee of correct output digits, and these values are not presented as certified intervals.

The first inverse-index correction need not improve an approximation at every small index. For example, at $m = 5$ the leading arctangent term happens to be unusually close, and its next term overshoots. The theorem concerns controlled errors as $m \rightarrow \infty$ and does not assert a globally enveloping expansion. The distinction between the inverse-logarithmic series and the

m	Approximate σ_m	Approximate ε_m
1	−2.151826895870	0.848173104130
2	−4.358674064874	0.641325935126
3	−6.454482934804	0.545517065196
5	−10.548589016994	0.451410983006
10	−20.640881124485	0.359118875515
50	−100.764467222239	0.235532777761
500	−1000.845421933140	0.154578066860

Table 3: Numerical illustrations of the proved real-zero classification. The decimals are diagnostic approximations.

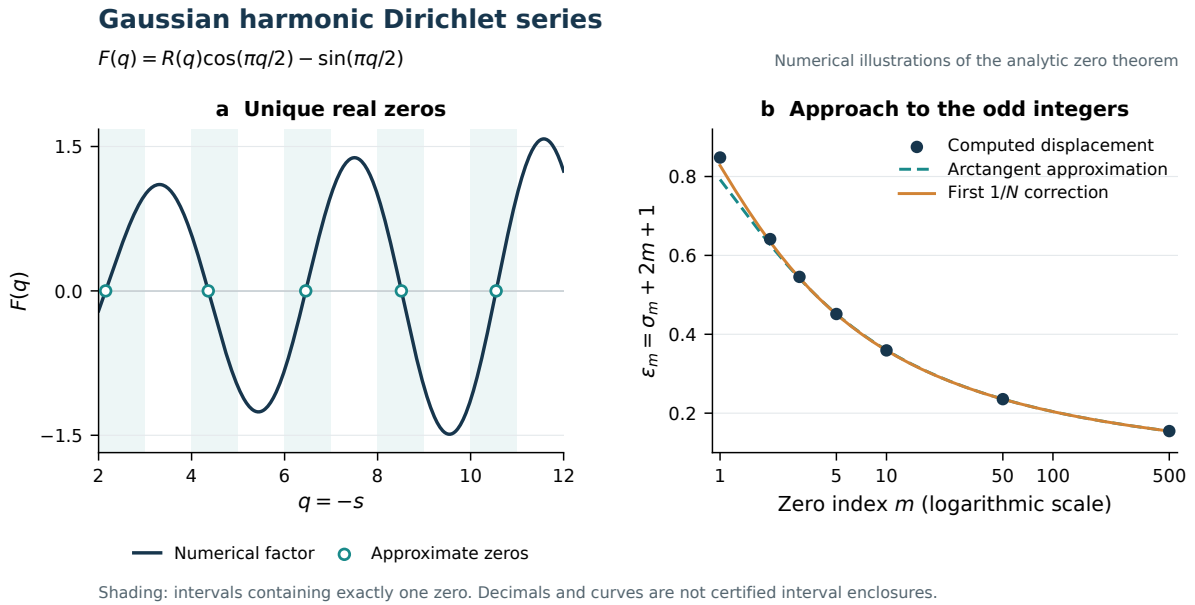


Figure 2: The normalized Fourier factor and the real-zero displacement. Zeros of $R(q)\cos(\pi q/2) - \sin(\pi q/2)$ give the real zeros of $S(-q)$ because its omitted prefactor is positive. The displacement plot compares numerical roots with the arctangent term and its first $1/N$ correction. These curves illustrate Theorems 6.1 and 7.2; they are not their proofs.

smaller $1/N$ sectors is also essential: retaining more logarithmic powers alone cannot resolve an algebraically small correction in N .

11.3 Certified sample cutoffs

The cutoff enclosures in Table 4 are exact. The old fixed-coefficient cutoffs in its last column are rounded diagnostics for comparison. In particular, the new coefficient can reduce the compact interval remaining for a separate zero count. That improvement does not itself supply signs on the compact interval.

n	Certified enclosure for B_n	Fixed $\lambda = e^{-n}$ cutoff
2	[1.47580064163065338930, 1.47580064163065338931]	1.5102804080
3	[2.27333271490894820528, 2.27333271490894820529]	2.3431706385
4	[3.57479611518613398335, 3.57479611518613398336]	3.6783274602
5	[5.69074542139040729581, 5.69074542139040729582]	5.8264887639
10	[63.21469931898564283555, 63.21469931898564283556]	63.5050628049

Table 4: Exact sample enclosures for the least uniform cutoff. Full optimizer, coefficient, and logarithmic-cutoff intervals are recorded in the accompanying JSON certificate.

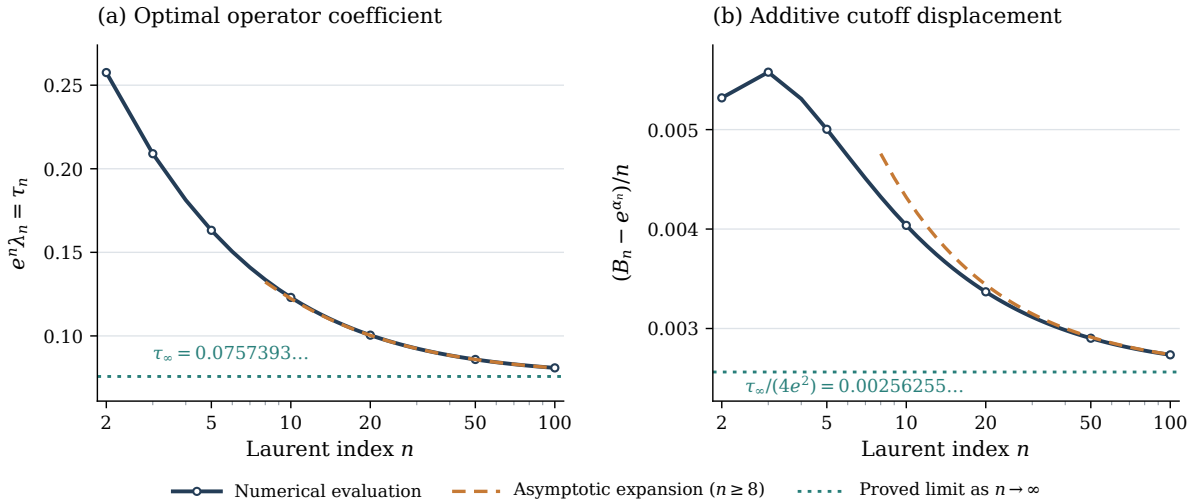


Figure 3: Optimal exterior-barrier parameters. Navy curves show numerical evaluations of the proved defining formulas; teal dotted lines show the limiting constants. Orange dashed curves, displayed for $n \geq 8$, use two inverse-index corrections for τ_n and the first correction for the normalized cutoff displacement. Numerical evaluation illustrates the analytic theorems; exact sample enclosures are recorded separately.

For the figure, the difference $B_n - e^{\alpha_n}$ is evaluated as $e^{\alpha_n} \expm1(\eta_n - \alpha_n)$ at high precision. Subtracting the two large exponentials directly would obscure the small displacement and can produce a misleading numerical comparison.

11.4 Replay and build instructions

The archive contains the complete expanded article.tex, the modular section sources, the compiled PDF, the data records, and all verification and plotting scripts. The standard-library exact checks and the symbolic/interval checks have separate entry points:

```
python3 code/verify_diagonal.py
python3 code/verify_gaussian_order.py --output data/gaussian_exact.json
python3 code/verify_optimal_barrier.py \
--output data/optimal_barrier_verification.json
```

The full numerical and quantitative commands, package requirements, and figure regeneration steps are in README.md. The optional numerical routines use mpmath, and the scientific figures use Matplotlib and SciPy. A requirements file records the environment used for the delivered run.

To regenerate the expanded TeX and PDF from the modular sources:

```
python3 code/assemble_article.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The distributed PDF was built from the same expanded source and visually checked. The build and verification records are included. The source snapshot identifies the baseline for later integration; the package does not modify the remote repository.

12 Manuscript corrections and integration

12.1 Replace the broad monotonicity suggestion by a quantified question

The final research chapter still says that monotonicity of higher zero branches in the deformation parameter is suggested by the first correction [5]. Read literally as a direction for all indices, that suggestion is obsolete once the incoming results are taken into account. The earlier boundary and bifurcation reports already give a turning lower branch at $(n, k) = (2, 1)$, while the global-phase report proves an increasing middle branch at $(3, 1)$ [6, 7, 8]. The positive initial velocity at $(n, k) = (4, 3)$ also appears in the formal-reductions report [9].

The additional correction supplied here is infinite in scope: Theorem 8.6 gives both directions at arbitrarily large indices on $k = n - 1$. A replacement paragraph should therefore ask for *effective derivative-order thresholds depending on the Laurent index*, the distribution of turning points below those thresholds, and the joint-index scales on which a uniform direction does emerge. It should not suggest that every higher branch decreases throughout the deformation.

One precise consequence is Corollary 8.7: if K_n guarantees decreasing motion for every branch and every $k \geq K_n$, then $K_n > n - 1$ for infinitely many n . This does not disprove the existence of K_n , does not show that K_n/n is unbounded, and does not identify a saturation threshold for the number of zeros. Those are different claims.

12.2 Record the exact endpoint in the optimized barrier

The solution of the incoming exterior-operator problem should be recorded in two parts. The least uniform cutoff for the summed family is attained and its inequality is strict when $\rho > 0$. The strict pointwise cutoff has the same infimum, but the infimum is not attained. At the optimal coefficient,

$$g'_n(x) + \lambda_n g_n(x) = 0 \iff x \in \{B_n, e^{n+r_n}\}, \quad x \geq B_n.$$

Consequently the strict summed theorem must not be extended unchanged to $\rho = 0$, to arbitrary one- or two-point positive measures, or to a claim that every individual summand is strictly negative. Theorem 10.1 and its positive-measure extension state the exact hypotheses. This is a qualification of the newly optimized result, not an assertion that the earlier fixed-coefficient theorem was false.

12.3 Keep integer special values separate from order-zero geometry

For reference, write $g_{a,b} = \text{Im Li}_{a,b}(i, 1)$ and $G = \beta(2)$. The remaining manuscript conjecture is

$$S(6) \stackrel{?}{=} \frac{722}{527}g_{6,1} + \frac{40}{527}g_{4,3} - \frac{128}{155}g_{2,5} + \frac{15191}{28569600}\pi^7 \\ - \frac{3}{124}G\zeta(5) - \frac{2373}{10540}\beta(4)\zeta(3) - 2\beta(6)\log 2. \quad (12.1)$$

This is quoted to identify the open target, not proposed as a new identity. The exact shuffle relation already in the manuscript is

$$g_{2,5} = 30g_{6,1} + 15g_{5,2} + 5g_{4,3} - \frac{15}{512}G\zeta(5) \\ + \frac{3}{16}\beta(4)\zeta(3) - \frac{11\pi^7}{368640}. \quad (12.2)$$

It proves equivalence of the two reported S_6 baskets [3]. It does not prove their residual vanishes.

Our Fourier–beta identity gives new exact representations of $S(6)$ by analytic continuation, but does not reduce it to the specified weight-seven period basket. Entirety and a complete real zero set likewise do not imply that reduction. The finite separating functional in the manuscript obstructs a particular restricted shuffle/stuffle presentation; it does not prove arithmetic independence. The successful S_4 certificate shows why a convergent octahedral relation and suitable higher-depth lifts are natural next ingredients [2, 3].

The incoming rejected S_8 vector is a useful methodological boundary: an interval containing zero is compatible with equality, but does not prove it; a certified interval excluding zero rejects the specific frozen vector, but does not prove independence of its basket. The proposed integration preserves those distinctions [5].

12.4 Notation and reusable proof components

The generalized secant numbers in Corollary 3.3 are positive at $\alpha = 1$. They differ by $(-1)^m$ from the signed Euler numbers sometimes defined through $\text{sech } t$. Their generating function, not the letter E alone, fixes the convention. The symbols S_p and $S(p)$ agree at positive integers. At negative integers $S(s)$ always denotes analytic continuation, since the original alternating Dirichlet series no longer converges.

The harmonic continuation and its Fourier identity can be integrated after the Gaussian harmonic sums. The diagonal theorem belongs after the elementary endpoint and fixed-index Lerch zero geometry. The optimized barrier should follow the incoming global-phase theorem, with its research question 2 marked resolved with the attainment qualification above. A suggested replacement paragraph and dependency map are included in the integration artifacts.

13 A proposed research programme

The results suggest several specific next problems. Each item below states what is missing and what would count as progress.

13.1 The order interpolation and its zeros

1. Nonreal zeros and growth of the entire function. Determine the growth order and directional indicator of $S(s)$, then locate and count its nonreal zeros. The real classification and the zero-free half-plane $\Re s \geq 3$ are starting constraints, not a classification in the complex plane. A useful first target is a rigorous argument-principle count in bounded rectangles using regularized Mellin integrals and controlled tails. A global conclusion would require uniform complex asymptotics.

2. Explicit finite-index errors for σ_m . Replace the implicit constants in Theorem 7.2 by computable bounds valid for every m beyond a stated threshold. The proof already reduces this to a sectorial digamma remainder and beta-series estimates. Combining those with a lower bound on the derivative of the tangent comparison would give certified intervals of predictable width without numerical root scanning.

3. Shifted denominators and higher harmonic depth. Study

$$S_a(s) = \sum_{n \geq 0} \frac{(-1)^n H_n}{(n+a)^s}, \quad a > 0,$$

and the elementary-symmetric harmonic sums already used by the manuscript. Their Mellin kernels are explicit, but the self-reflecting choice $a = 1/2$ is special: its Fourier transform gives the positive moment ratio used here. Determine which shifts preserve a comparable sign or total-positivity structure. It would be premature to assert the same real zero pattern for every shift.

4. Sharper moment-ratio inequalities. The estimate $R'(q) < \sqrt{181/27}/q$ is intentionally coarse. Find the best constant or a useful pointwise comparison, and determine whether R' decreases on $[2, \infty)$. The proof establishes neither concavity nor complete monotonicity of R . A refined covariance inequality may be more efficient than additional moment bounds.

13.2 Joint-index Lerch geometry

5. Prove or refute the accessible-singularity conjecture. The exact inverse equation for u_A makes its finite critical values explicit, but those values belong to different analytic sheets. For $A = 2$, prove which critical points are accessible from the germ at zero, identify all singularities on its convergence circle, and establish the continuation domain needed for coefficient transfer. This would turn the quantitative conjecture into an asymptotic theorem. Numerical radial continuation and agreement of many coefficients are evidence, not a substitute for that domain argument.

6. Density and gaps of the two velocity signs. Theorem 8.6 proves infinitude of both signs without a density or a maximum gap. A dominant-conjugate-pair theorem should give an oscillatory law, but a density statement also depends on the arithmetic and possible resonances of its phase. For algebraic $A > 1$, the finite Stirling formula already excludes a zero velocity; it does not bound how small a nonzero velocity may be.

7. Effective monotonicity thresholds. Determine the growth of the least K_n for which all zero branches decrease for $k \geq K_n$. The present lower constraint $K_n > n - 1$ along infinitely many indices should be combined with the existing fixed- n concentration estimates. A linear upper bound, a sharp transition ratio, or a proof that no linear bound is possible would each be meaningful outcomes; none follows merely from the diagonal sign theorem.

8. Central root splitting for $n = k + r$. For fixed $r > 1$, the elementary root at $a = 1$ has multiplicity r . Its natural perturbation can involve fractional powers of ρ . Derive a uniform local normal form, determine the number of real branches it produces, and compare it with the existing expansions near positive Bessel zeros [10]. The central root and positive Bessel roots should not be treated as interchangeable regimes.

9. Beyond first-order velocity. Find generating functions for higher ρ derivatives of $a_n^{(h)}(\rho)$ and for the size of the neighborhood in which the first derivative determines its motion. Small nonzero v_n can force second-order terms to become relevant unusually early. This is a concrete way to connect the local oscillation theorem to actual turning points in the full deformation.

13.3 Exterior barriers and remaining phase questions

10. Fixed-parameter and variable-coefficient barriers. Theorem 10.1 optimizes a constant coefficient uniformly for arbitrarily small positive ρ . For a fixed positive ρ , the tail sum may allow a smaller cutoff. Determine the optimum as a function of ρ , or allow a controlled positive function $\lambda(a)$. Such an extension must specify its admissible class; otherwise a best-operator question is not a well-defined optimization problem.

11. Existence inside the exterior uniqueness region. The optimal barrier proves at most one exterior zero and a downward crossing. It does not guarantee a zero. Prove signs at B_n for useful ranges of (n, ρ) , perhaps using a small set of exact separators and geometric-weight inequalities. This would turn a one-crossing bound into an all-index existence and uniqueness theorem.

12. Strict span unimodality at index three. The incoming report already proves the exact unit maximum and its nondegeneracy, but its stronger conjecture of strict increase before the maximum and strict decrease afterwards remains open [8]. Additional control of the averaged outer slope is required; the sign of the shifted-root identity alone does not settle this shape question.

13.4 Arithmetic reductions and formal verification

13. A weight-seven mixed certificate. For (12.1), seek a convergent octahedral relation whose image is nonzero under the known restricted separating functional. Then lift and eliminate depth-three or higher terms with exact rational rows. The useful output is a short certificate with an independently specified word convention and regularization, rather than another decimal relation fit. A failed search in a finite row space should be recorded as a limitation of that row space.

14. Formalize the finite and analytic interfaces separately. The Stirling-polynomial recurrence, cubic factorization, rational interval operations, and negative-integer recurrences are finite targets suited to proof-assistant formalization. The subsequent analytic targets are Mellin subtraction, the one-dimensional variance inequality, the positive-coefficient argument, and the implicit zero comparison. The current package provides ordinary proofs and executable checks; it does not claim a completed Lean or other proof-assistant verification.

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